Vector Space Lecture Notes

Compiled for Exam Preparation

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Contents

1 Lecture 01: Vector Space - Concept and Definition

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 01 of 27 **Video**: https://www.youtube.com/watch?v=_6oRqxY6O5w

1.1 Prerequisites

Before we define a vector space, we need to understand two key types of operations. These operations tell us how elements combine.

1.1.1 Internal Composition

Internal composition is an operation where you combine two elements from the same set, and the result stays in that set.

Imagine a non-empty set A. Take two elements α and β from A. If you add them together using an operation $+$, and the new element γ also belongs to A, then $+$ is an internal composition for A.

$$\alpha + \beta = \gamma \in A$$

1.1.2 External Composition

External composition involves two different sets. You take an element from one set, apply an operation using an element from a second set, and the result belongs to the first set.

Let A be our main set and B be another set (which we will later call a field). Take an element α from A and an element a from B. If you multiply them and the result β belongs to A, then this multiplication is an external composition for A.

$$a \cdot \alpha = \beta \in A$$

This means the operation is defined using elements from B, but the output always lands back in A.

1.2 What is a Vector Space?

Now we can formally define a vector space. To build a vector space, we need two sets: 1. A set V, whose elements are called **vectors**. 2. A field F (usually real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$), whose elements are called **scalars**.

The set V is called a **Vector Space** over the field F if it satisfies certain conditions under two operations: vector addition (internal) and scalar multiplication (external). We denote this as $V(F)$.

Here are the conditions $V(F)$ must satisfy:

1.2.1 1. Abelian Group under Vector Addition

The set V must be an abelian group with respect to the internal composition $+$. This means: * **Closure**: Adding two vectors gives a vector in V. * **Associativity**: $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$. * **Identity**: There is a zero vector $\bar{0}$ such that $\alpha + \bar{0} = \alpha$. * **Inverse**: Every vector α has an additive inverse $-\alpha$ such that $\alpha + (-\alpha) = \bar{0}$. * **Commutativity**: $\alpha + \beta = \beta + \alpha$.

1.2.2 2. Closure under Scalar Multiplication

The set V must be closed under external composition. If you multiply a scalar from F with a vector from V, the result must be a vector in V.

For all $a \in F$ and for all $\alpha \in V$:

$$a \cdot \alpha \in V$$

1.2.3 3. Distribution and Associative Laws

The vector space must follow these rules connecting scalars and vectors:

- **Distributive Law over Vector Addition**:

$$a \cdot (\alpha + \beta) = a \cdot \alpha + a \cdot \beta \quad \text{for all } a \in F, \alpha, \beta \in V$$

- **Distributive Law over Scalar Addition**:

$$(a + b) \cdot \alpha = a \cdot \alpha + b \cdot \alpha \quad \text{for all } a, b \in F, \alpha \in V$$

- **Associative Law for Scalar Multiplication**:

$$(ab) \cdot \alpha = a \cdot (b \cdot \alpha) \quad \text{for all } a, b \in F, \alpha \in V$$

- **Identity Scalar Multiplication**:

$$1 \cdot \alpha = \alpha \quad \text{for all } \alpha \in V$$

`where $1$ is the multiplicative identity of the field $F$.`

If a set V combined with a field F meets all these rules, we proudly call V a vector space. The vectors in V do not have to be traditional directed lines. They can be points on a plane, matrices, or even polynomials.

1.3 Example: The 2D Plane as a Vector Space

Let us verify that the 2D Cartesian plane is a vector space over the field of real numbers.

- **Vectors**:

$$V = \mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

- **Scalars**: $F = \mathbb{R}$

Let us check the conditions.

1.3.1 Check 1: Is an Abelian Group?

Take two vectors (x_1, y_1) and (x_2, y_2) from V. * **Closure**:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

. Since the sum of two real numbers is a real number, this new tuple is in V. * **Identity**: The element $(0, 0)$ works as the identity because $(x, y) + (0, 0) = (x, y)$. * **Inverse**: The inverse of (x, y) is $(-x, -y)$ because $(x, y) + (-x, -y) = (0, 0)$. * **Commutativity**:

$$(x_1, y_1) + (x_2, y_2) = (x_2, y_2) + (x_1, y_1)$$

is naturally true for real numbers.

1.3.2 Check 2: Closure under Scalar Multiplication

Take a scalar a from $\mathbb{R}$ and a vector (x, y) from V.

$$a(x, y) = (ax, ay)$$

Since ax and ay are real numbers, the result (ax, ay) is in V.

1.3.3 Check 3: Distributive Laws

Let us prove the first distributive law. We want to show $a(\alpha + \beta) = a\alpha + a\beta$. Let

$$\alpha = (x_1, y_1)$$

and

$$\beta = (x_2, y_2).$$

$$a(\alpha + \beta) = a((x_1, y_1) + (x_2, y_2))$$

First, add the vectors inside the bracket:

$$= a(x_1 + x_2, y_1 + y_2)$$

Now, multiply the scalar a inside:

$$= (a(x_1 + x_2), a(y_1 + y_2))$$

Expand using basic algebra:

$$= (ax_1 + ax_2, ay_1 + ay_2)$$

Split this back into two tuples:

$$= (ax_1, ay_1) + (ax_2, ay_2)$$

Factor out the scalar a:

$$= a(x_1, y_1) + a(x_2, y_2)$$

Which gives us our final result:

$$= a\alpha + a\beta$$

Let us prove the second distributive law. We want to show $(a + b)\alpha = a\alpha + b\alpha$. Let $\alpha = (x_1, y_1)$.

$$(a + b)\alpha = (a + b)(x_1, y_1)$$

$$= ((a + b)x_1, (a + b)y_1)$$

$$= (ax_1 + bx_1, ay_1 + by_1)$$

$$= (ax_1, ay_1) + (bx_1, by_1)$$

$$= a(x_1, y_1) + b(x_1, y_1)$$

$$= a\alpha + b\alpha$$

Next, let us prove scalar associativity. We want to show $(ab)\alpha = a(b\alpha)$.

$$(ab)\alpha = (ab)(x_1, y_1)$$

$$= ((ab)x_1, (ab)y_1)$$

$$= (a(bx_1), a(by_1))$$

$$= a(bx_1, by_1)$$

$$= a(b(x_1, y_1))$$

$$= a(b\alpha)$$

Finally, identity scalar multiplication:

$$1 \cdot \alpha = 1(x_1, y_1)$$

$$= (1 \cdot x_1, 1 \cdot y_1)$$

$$= (x_1, y_1)$$

$$= \alpha$$

Since all conditions are met, $\mathbb{R}^2$ is a vector space over $\mathbb{R}$.

1.3.4 Other Common Examples

- **Example 2:** The set of complex numbers $\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$ is a vector space over the field of real numbers $\mathbb{R}$.
- **Example 3:** The set of 3D coordinates $V = \mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ is a vector space over the field of real numbers $\mathbb{R}$.

1.4 Key Takeaways

- An **internal composition** maps elements within the same set.
- An **external composition** maps an element from a scalar field and an element from a vector set back into the vector set.
- A **Vector Space** is a set of vectors that forms an abelian group under addition and is closed under scalar multiplication with elements from a field.
- The vectors must obey specific distributive and associative rules connecting vector addition and scalar multiplication.

1.5 What Comes Next

2 Lecture 02: Vector Space - General Properties

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 02 of 27 **Video**: https://www.youtube.com/watch?v=yzuoyA2EJPA

2.1 Prerequisites

We already know that a vector space $V(F)$ requires a set of vectors V and a set of scalars F. The vectors form an abelian group under addition. They also follow specific distribution and closure rules when multiplied by scalars from F.

2.2 Notation: Zero Vector vs. Zero Scalar

Before we prove anything, we must be clear about zeros. * We use $\bar{0}$ to represent the **zero vector** in V. * We use 0 to represent the **zero scalar** in F.

This distinction is important because vectors and scalars are different types of objects.

2.3 General Properties of Vector Spaces

Let $V(F)$ be a vector space. The following six general properties always hold true.

1. $a\bar{0} = \bar{0} \quad \text{for all } a \in F$
2. $0\alpha = \bar{0} \quad \text{for all } \alpha \in V$
3. $a(-\alpha) = -(a\alpha) \quad \text{for all } a \in F, \alpha \in V$
4. $(-a)\alpha = -(a\alpha) \quad \text{for all } a \in F, \alpha \in V$
5. $a(\alpha - \beta) = a\alpha - a\beta \quad \text{for all } a \in F, \alpha, \beta \in V$
6. If $a\alpha = \bar{0}$, then either $a = 0$ or $\alpha = \bar{0}$.

Let us prove each property step by step. A common trick in these proofs is knowing where to start. We often start with an identity equation and multiply both sides by a scalar or vector.

2.3.1 Proof of Property 1

We want to multiply a scalar by the zero vector. So we start with a known fact about the zero vector.

We know that adding the zero vector to itself gives the zero vector:

$$\bar{0} + \bar{0} = \bar{0}$$

Multiply both sides by scalar a:

$$a(\bar{0} + \bar{0}) = a\bar{0}$$

Apply the distributive law:

$$a\bar{0} + a\bar{0} = a\bar{0}$$

Now, add $\bar{0}$ to the right side. Adding the zero vector does not change a vector.

$$a\bar{0} + a\bar{0} = a\bar{0} + \bar{0}$$

By the left cancellation law in the vector group $(V, +)$, we cancel $a\bar{0}$ from the left of both sides:

$$a\bar{0} = \bar{0}$$

This completes the first proof.

2.3.2 Proof of Property 2

Here we want to multiply the zero scalar by a vector. We start with the zero scalar. We know that adding zero to zero gives zero:

$$0 + 0 = 0$$

Multiply both sides by vector α from the right:

$$(0 + 0)\alpha = 0\alpha$$

Apply the distributive law:

$$0\alpha + 0\alpha = 0\alpha$$

Add the zero vector $\bar{0}$ to the right side:

$$0\alpha + 0\alpha = 0\alpha + \bar{0}$$

Again, apply the left cancellation law to remove 0α from both sides:

$$0\alpha = \bar{0}$$

This proves the second property.

2.3.3 Proof of Property 3

We start with a vector and its additive inverse. We know they sum to the zero vector.

$$\alpha + (-\alpha) = \bar{0}$$

Multiply both sides by scalar a:

$$a[\alpha + (-\alpha)] = a\bar{0}$$

Apply the distributive law on the left. On the right, use Property 1 ($a\bar{0} = \bar{0}$):

$$a\alpha + a(-\alpha) = \bar{0}$$

We want to isolate $a(-\alpha)$. So we add the additive inverse $-(a\alpha)$ to both sides:

$$-(a\alpha) + [a\alpha + a(-\alpha)] = -(a\alpha) + \bar{0}$$

Use associativity to group the first two terms:

$$[-(a\alpha) + a\alpha] + a(-\alpha) = -(a\alpha)$$

The terms in the brackets sum to the zero vector $\bar{0}$:

$$\bar{0} + a(-\alpha) = -(a\alpha)$$

Adding the zero vector leaves the vector unchanged, giving us our final result:

$$a(-\alpha) = -(a\alpha)$$

2.3.4 Proof of Property 4

This proof is very similar to Property 3. But here we start with scalar addition. We know a scalar and its additive inverse sum to the zero scalar:

$$a + (-a) = 0$$

Multiply both sides by vector α:

$$[a + (-a)]\alpha = 0\alpha$$

Distribute on the left, and use Property 2 on the right ($0\alpha = \bar{0}$):

$$a\alpha + (-a)\alpha = \bar{0}$$

Add $-(a\alpha)$ to both sides:

$$-(a\alpha) + [a\alpha + (-a)\alpha] = -(a\alpha) + \bar{0}$$

Group using associativity:

$$[-(a\alpha) + a\alpha] + (-a)\alpha = -(a\alpha)$$

The terms in brackets sum to $\bar{0}$:

$$\bar{0} + (-a)\alpha = -(a\alpha)$$

Which simplifies to:

$$(-a)\alpha = -(a\alpha)$$

2.3.5 Proof of Property 5

We only have distributive laws for addition, not subtraction. But we can write vector subtraction as adding an inverse.

Start with the left side:

$$a(\alpha - \beta) = a[\alpha + (-\beta)]$$

Now we can apply the standard distributive law:

$$= a\alpha + a(-\beta)$$

From Property 3, we know $a(-\beta) = -(a\beta)$. Substitute this in:

$$= a\alpha + [-(a\beta)]$$

Which we simply write as subtraction:

$$= a\alpha - a\beta$$

2.3.6 Proof of Property 6: If, then or

This property has two parts. If the product of a scalar and a vector is zero, one of them must be zero.

Part A: Assume $a \neq 0$ If the scalar a is not zero, then it must have a multiplicative inverse a^{-1} in the field F. Start with our given equation:

$$a\alpha = \bar{0}$$

Multiply both sides by a^{-1}:

$$a^{-1}(a\alpha) = a^{-1}\bar{0}$$

Use associativity for scalars on the left, and Property 1 on the right:

$$(a^{-1}a)\alpha = \bar{0}$$

Since

$$a^{-1}a = 1:$$

$$1 \cdot \alpha = \bar{0}$$

By the vector space rules, $1 \cdot \alpha = \alpha$. Thus:

$$\alpha = \bar{0}$$

So if $a \neq 0$, then α must be $\bar{0}$.

Part B: Assume $\alpha \neq \bar{0}$ We want to prove $a = 0$. We will use proof by contradiction. Let us assume $a \neq 0$. If $a \neq 0$, then a^{-1} exists in F. Starting again with:

$$a\alpha = \bar{0}$$

Multiply by a^{-1} on the left:

$$a^{-1}(a\alpha) = a^{-1}\bar{0}$$

Following the exact same steps as above, this simplifies to:

$$1 \cdot \alpha = \bar{0}$$

$$\alpha = \bar{0}$$

But this result contradicts our assumption that $\alpha \neq \bar{0}$. Because assuming $a \neq 0$ leads to a contradiction, our assumption must be false. Therefore, a must be 0.

2.4 Key Takeaways

- Multiplying any scalar by the zero vector gives the zero vector.
- Multiplying the zero scalar by any vector gives the zero vector.
- Negative signs factor out smoothly: $a(-\alpha) = (-a)\alpha = -(a\alpha)$.
- Scalar multiplication distributes over vector subtraction.
- If the product of a scalar and a vector is the zero vector, it is guaranteed that at least one of them is zero.

2.5 What Comes Next

3 Lecture 03: Vector Space - Vector Subspace

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 03 of 27 **Video**: https://www.youtube.com/watch?v=5LCO26C-ggo

3.1 Prerequisites

We know the ten axioms required for a set to be a vector space $V(F)$. Checking all ten rules every time is tedious. This lecture introduces subspaces and a shortcut theorem for proving them. Think of a subspace just like a subgroup in group theory.

3.2 Definition of a Vector Subspace

Suppose we have a vector space $V(F)$. Let W be a subset of V ($W \subseteq V$).

We say that W is a **vector subspace** of V if W is itself a vector space under the exact same operations defined on V.

To be a subspace, W must be a working vector space on its own. While this is the formal definition, we rarely check all vector space properties to prove a subset is a subspace. Instead, we use a single, powerful theorem.

3.3 The One-Step Subspace Theorem

This theorem gives us a necessary and sufficient condition. It is the primary tool used to solve numerical questions about subspaces.

Theorem Statement: A non-empty subset W of a vector space $V(F)$ is a subspace of $V(F)$ if and only if for all scalars $a, b \in F$ and all vectors $\alpha, \beta \in W$, their linear combination belongs to W:

$$a\alpha + b\beta \in W$$

Because this is an "if and only if" theorem, the proof has two parts. We must show the condition is necessary (it must hold if W is a subspace) and sufficient (if it holds, W is guaranteed to be a subspace).

3.3.1 Part 1: The Condition is Necessary

Let us assume W is a vector subspace of $V(F)$.

Since W is a subspace, it acts as a vector space in its own right. This means W must be closed under scalar multiplication. For any $\alpha \in W$ and $a \in F$:

$$a\alpha \in W$$

Similarly, for another vector $\beta \in W$ and scalar $b \in F$:

$$b\beta \in W$$

Because W is a subspace, it must also be closed under vector addition. We can add our two new vectors $a\alpha$ and $b\beta$ together, and the sum must remain in W:

$$a\alpha + b\beta \in W$$

This proves the condition is necessary.

3.3.2 Part 2: The Condition is Sufficient

Now we prove the reverse. We assume only that W is a subset and the condition holds: For all $a, b \in F$ and $\alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W$. Let's call this Condition (1).

We must prove W is a full vector space. We do this by choosing clever values for the scalars a and b to check off the vector space requirements.

Step 1: Existence of Additive Inverse Put $a = 0$ and $b = -1$ into Condition (1):

$$0 \cdot \alpha + (-1) \cdot \beta = \bar{0} - \beta = -\beta \in W$$

This shows that for every vector β in W, its additive inverse $-\beta$ is also in W.
Step 2: Existence of Zero Vector (Identity) Put $a = 0$ and $b = 0$ into Condition (1):

$$0 \cdot \alpha + 0 \cdot \beta = \bar{0} + \bar{0} = \bar{0} \in W$$

This proves the zero vector is in W.
Step 3: Closure Under Vector Addition Put $a = 1$ and $b = 1$ into Condition (1):

$$1 \cdot \alpha + 1 \cdot \beta = \alpha + \beta \in W$$

This proves W is closed under vector addition.
Since W is closed under addition, has a zero vector, and contains additive inverses, $(W, +)$ is an abelian group. The commutative and associative laws for addition hold automatically because the elements in W are also in V, where those laws already work.
Step 4: Closure Under Scalar Multiplication Put $a = 0$ (and leave b as any scalar) into Condition (1):

$$0 \cdot \alpha + b \cdot \beta = \bar{0} + b\beta = b\beta \in W$$

This proves W is closed under scalar multiplication.
Finally, the scalar distribution laws hold automatically in W because they are inherited from the parent space V. Since $(W, +)$ is an abelian group, is closed under scalar multiplication, and inherits all distribution laws, W is a vector space.
Therefore, W is a subspace of $V(F)$. The proof is complete.

3.4 Numerical Examples

How do we use this theorem in practice? We pick two arbitrary vectors from the subset and two scalars, form the linear combination $a\alpha + b\beta$, and check if the result matches the structural rule of the subset.

3.4.1 Example 1: A Subspace of 3D Space

Question: Show that the set $W = \{(x, y, 0) \mid x, y \in F\}$ is a subspace of $V_3(F)$.
Solution: The subset W contains 3-tuples where the third coordinate is always exactly zero, and the first two coordinates are scalars from F.
Let us pick two vectors from W:

$$\alpha = (x_1, y_1, 0) \quad \text{and} \quad \beta = (x_2, y_2, 0)$$

And choose two scalars $a, b \in F$.
We form the linear combination:

$$a\alpha + b\beta = a(x_1, y_1, 0) + b(x_2, y_2, 0)$$

Multiply the scalars into the tuples:

$$= (ax_1, ay_1, 0) + (bx_2, by_2, 0)$$

Add the tuples coordinate by coordinate:

$$= (ax_1 + bx_2, ay_1 + by_2, 0 + 0)$$
$$= (ax_1 + bx_2, ay_1 + by_2, 0)$$

Now, we check the structure of our result. Since F is a field, the sums and products of scalars are also scalars. So, $(ax_1 + bx_2)$ is a scalar in F, and $(ay_1 + by_2)$ is a scalar in F.
The resulting tuple has two scalars in the first two positions, and a zero in the third position. This perfectly matches the rule for W.

$$a\alpha + b\beta \in W$$

Therefore, W is a subspace of $V_3(F)$.

3.4.2 Example 2: A Subspace of Matrices

Question: Show that the set

$$W = \left\{ \begin{pmatrix} x & y \\ -y & x \end{pmatrix} \mid x, y \in F \right\}$$

is a subspace of $M_2(F)$.

Solution: $M_2(F)$ is the vector space of all 2×2 matrices. The subset W has a specific rule: the diagonal elements must match each other, and the off-diagonal elements must be negatives of each other.

Pick two matrices from W:

$$\alpha = \begin{pmatrix} x_1 & y_1 \\ -y_1 & x_1 \end{pmatrix} \quad \text{and} \quad \beta = \begin{pmatrix} x_2 & y_2 \\ -y_2 & x_2 \end{pmatrix}$$

And pick scalars $a, b \in F$.

Form the combination:

$$a\alpha + b\beta = a \begin{pmatrix} x_1 & y_1 \\ -y_1 & x_1 \end{pmatrix} + b \begin{pmatrix} x_2 & y_2 \\ -y_2 & x_2 \end{pmatrix}$$

Multiply the scalars inside:

$$= \begin{pmatrix} ax_1 & ay_1 \\ -ay_1 & ax_1 \end{pmatrix} + \begin{pmatrix} bx_2 & by_2 \\ -by_2 & bx_2 \end{pmatrix}$$

Add the matrices element by element:

$$= \begin{pmatrix} ax_1 + bx_2 & ay_1 + by_2 \\ -ay_1 - by_2 & ax_1 + bx_2 \end{pmatrix}$$

Factor out the negative sign in the bottom-left corner to check the structure:

$$= \begin{pmatrix} ax_1 + bx_2 & ay_1 + by_2 \\ -(ay_1 + by_2) & ax_1 + bx_2 \end{pmatrix}$$

Look at the structure. The diagonal elements $(ax_1 + bx_2)$ are equal. The top-right element is $(ay_1 + by_2)$, and the bottom-left element is its negative. This perfectly matches the rule for W.

$$a\alpha + b\beta \in W$$

Therefore, W is a subspace of $M_2(F)$.

3.5 Key Takeaways

- A **vector subspace** is a subset that behaves as a full vector space under the parent space's operations.
- To prove a subset W is a subspace, we use the one-step test: $a\alpha + b\beta \in W$.
- We can prove the zero vector, additive inverses, and closure exist by cleverly substituting 0, 1, and -1 for the scalars a and b in the one-step test.
- For numerical problems, always take two elements structured like W, combine them with scalars, and verify the resulting expression shares the exact same structure.

3.6 What Comes Next

4 Lecture 04: Vector Space - Intersection and Union of Vector Subspaces

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 04 of 27 **Video**: https://www.youtube.com/watch?v=QZi8OqYSLgU

4.1 Prerequisites

We know that a subset W of a vector space $V(F)$ is a subspace if it behaves as a vector space itself. The one-step test says W is a subspace if for all scalars a, b and vectors α, β in W, the combination $a\alpha + b\beta$ is also in W.

Today we will use this theorem to see what happens when we combine two subspaces using intersections and unions.

4.2 Intersection of Two Subspaces

When we take the intersection of two vector subspaces, we look at all the vectors they share. We will prove that this intersection forms a brand new subspace.

Theorem: Prove that the intersection of two vector subspaces of a vector space is also a vector subspace.

Proof: Let W_1 and W_2 be two vector subspaces of $V(F)$. We want to prove that $W_1 \cap W_2$ is a vector subspace.

First, we must show that the intersection is not an empty set. Since W_1 and W_2 are both subspaces, they act as vector spaces under addition. This means the additive identity, which is the zero vector $\bar{0}$, must exist in both. * $\bar{0} \in W_1$ * $\bar{0} \in W_2$

Since the zero vector is in both sets, it must be in their intersection:

$$\bar{0} \in W_1 \cap W_2$$

This proves that $W_1 \cap W_2$ is non-empty.

Now we will use the one-step subspace theorem. Let us choose two arbitrary vectors α and β from the intersection:

$$\alpha, \beta \in W_1 \cap W_2$$

Since these vectors are in the intersection, they belong to both sets individually: * $\alpha, \beta \in W_1$ * $\alpha, \beta \in W_2$

Now let us take any two scalars $a, b \in F$.

Since W_1 is a subspace, and $\alpha, \beta \in W_1$:

$$a\alpha + b\beta \in W_1$$

Since W_2 is a subspace, and $\alpha, \beta \in W_2$:

$$a\alpha + b\beta \in W_2$$

Because the vector sum $a\alpha + b\beta$ belongs to both W_1 and W_2, it must belong to their intersection:

$$a\alpha + b\beta \in W_1 \cap W_2$$

Thus, by the one-step subspace theorem, the intersection $W_1 \cap W_2$ is a vector subspace of $V(F)$.

4.3 Union of Two Subspaces

The union of two sets contains all elements from the first set and all elements from the second set. Unlike the intersection, the union of two subspaces is **not always** a subspace. It only forms a subspace under a specific condition.

Theorem: If W_1 and W_2 are two subspaces of $V(F)$, prove that the union $W_1 \cup W_2$ is a vector subspace of $V(F)$ if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Because of the "if and only if" condition, we must prove this in both directions.

4.3.1 Part 1: The condition is sufficient

We start by assuming that one subspace is contained in the other. Let $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

- If $W_1 \subseteq W_2$, then all elements of W_1 are already in W_2. Thus, their union is simply the larger set:

$$W_1 \cup W_2 = W_2$$

. Since W_2 is given as a subspace, the union is a subspace. * If $W_2 \subseteq W_1$, then

$$W_1 \cup W_2 = W_1$$

. Since W_1 is given as a subspace, the union is a subspace.

In both cases, the union is just one of the original subspaces. So $W_1 \cup W_2$ is a subspace.

4.3.2 Part 2: The condition is necessary

Now we prove the reverse direction. We assume that the union $W_1 \cup W_2$ is a vector subspace, and we must prove that one set is contained in the other ($W_1 \subseteq W_2$ or $W_2 \subseteq W_1$).

We will prove this by contradiction. We assume the opposite is true: neither set is contained in the other. Suppose $W_1 \nsubseteq W_2$ and $W_2 \nsubseteq W_1$.

What does this mean? * Since W_1 is not contained in W_2, there is some vector α that belongs to W_1 but not W_2. Let's call this condition (1). * Since W_2 is not contained in W_1, there is some vector β that belongs to W_2 but not W_1. Let's call this condition (2).

Since $\alpha \in W_1$ and $\beta \in W_2$, both vectors must belong to their union:

$$\alpha, \beta \in W_1 \cup W_2$$

We assumed that $W_1 \cup W_2$ is a subspace. This means it must be closed under vector addition. So, the sum of these two vectors must be in the union:

$$\alpha + \beta \in W_1 \cup W_2$$

If this sum is in the union, it must belong to either W_1 or W_2. Let us test both possibilities.

Case A: Suppose $\alpha + \beta \in W_1$. We know W_1 is a subspace. We also know $\alpha \in W_1$. Because W_1 acts as a vector group, the additive inverse $-\alpha$ is also in W_1. Since both $(\alpha + \beta)$ and $-\alpha$ are in W_1, their sum must be in W_1:

$$(\alpha + \beta) + (-\alpha) \in W_1$$

$$\beta \in W_1$$

But this directly contradicts condition (2), which states $\beta \notin W_1$.

Case B: Suppose $\alpha + \beta \in W_2$. We know W_2 is a subspace, and $\beta \in W_2$. Thus, the additive inverse $-\beta \in W_2$. Since both $(\alpha + \beta)$ and $-\beta$ are in W_2, their sum must be in W_2:

$$(\alpha + \beta) + (-\beta) \in W_2$$

$$\alpha \in W_2$$

But this directly contradicts condition (1), which states $\alpha \notin W_2$.

In both cases, we hit a solid contradiction. Our assumption that neither set was contained in the other must be wrong. Therefore, it is absolutely necessary that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. The theorem is proven.

4.4 Key Takeaways

- The **intersection** of two vector subspaces is always a vector subspace.
- The **union** of two vector subspaces is only a subspace if one is entirely contained within the other.
- Proof by contradiction is a powerful tool when working with unions.

4.5 What Comes Next

5 Lecture 05: Vector Space - Concept of Linear Sum, Direct Sum & Linear Span

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 05 of 27 **Video**: https://www.youtube.com/watch?v=kSyYgXvBpQQ

5.1 Prerequisites

Before we dive into creating sums from different subspaces, we need to know how to combine vectors using scalars. This is the foundation for creating sets like the linear span.

5.2 Linear Combination of Vectors

If we are given a set of vectors, we can mix them together using scalars to create a new vector. This process is called forming a linear combination.

Let

$$\alpha_1, \alpha_2, \dots, \alpha_n$$

be vectors in a vector space $V(F)$. Let $a_1, a_2, \dots, a_n \in F$ be scalars from our field.

The **linear combination** of these vectors is defined as:

$$\sum_{i=1}^{n} a_i \alpha_i = a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n = \alpha$$

Because vector spaces are closed under scalar multiplication and vector addition, the result α is always a new vector that belongs to $V(F)$.

5.3 Linear Span of a Set

What happens if we take a set of vectors and form *every possible* linear combination using every possible scalar from the field? We generate a massive set of new vectors. This new set is called the linear span.

Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_n\} \subseteq V(F)$$

be a set of vectors.

The **linear span** of S, denoted by $L(S)$ or $L[S]$, is the set of all possible linear combinations of the vectors in S.

$$L(S) = \left\{ \sum_{i=1}^{n} a_i \alpha_i \mid a_i \in F \text{ and } \alpha_i \in S \right\}$$

Since every linear combination gives us a vector in $V(F)$, the linear span is itself a subset of the vector space: $L(S) \subseteq V(F)$. The linear span represents the entire space that can be built or "reached" using just the vectors in S.

5.4 Linear Sum of Two Subspaces

Now let us look at combining two different subspaces. We already learned that the union of two subspaces is usually not a subspace. However, we can create a new subspace by adding them together element by element.

Let W_1 and W_2 be two vector subspaces of $V(F)$.

The **linear sum** of W_1 and W_2 is denoted by $W_1 + W_2$ and is defined as:

$$W_1 + W_2 = \{\alpha_1 + \beta_1 \mid \alpha_1 \in W_1 \text{ and } \beta_1 \in W_2\}$$

In simple terms, you pick one vector from W_1 and one vector from W_2, add them together, and place the result in the new set.

5.4.1 Proving the Linear Sum is a Subspace

We can easily prove that $W_1 + W_2$ is a vector subspace of $V(F)$ using the one-step test.
Let us pick two vectors α and β from $W_1 + W_2$. By definition, they must be formed by sums: *

$$\alpha = \alpha_1 + \alpha_2$$

- $$\beta = \beta_1 + \beta_2$$

where $\alpha_1, \beta_1 \in W_1$ and $\alpha_2, \beta_2 \in W_2$.
Now, let $a, b \in F$ be any two scalars. We check their linear combination:

$$a\alpha + b\beta = a(\alpha_1 + \alpha_2) + b(\beta_1 + \beta_2)$$

Distribute the scalars:

$$= (a\alpha_1 + a\alpha_2) + (b\beta_1 + b\beta_2)$$

Rearrange to group the W_1 terms together and the W_2 terms together:

$$= (a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2)$$

Because W_1 is a subspace, the combination $(a\alpha_1 + b\beta_1)$ belongs to W_1. Because W_2 is a subspace, the combination $(a\alpha_2 + b\beta_2)$ belongs to W_2.
Thus, our new vector $a\alpha + b\beta$ is written exactly as a sum of a vector in W_1 and a vector in W_2. Therefore:

$$a\alpha + b\beta \in W_1 + W_2$$

This proves that the linear sum $W_1 + W_2$ is a vector subspace of $V(F)$.

5.5 Direct Sum of Two Subspaces

A direct sum is a special, strict version of the linear sum.
Let W_1 and W_2 be two subspaces of $V(F)$. The **direct sum** of W_1 and W_2 is denoted by $W_1 \oplus W_2$.
It is defined identically to the linear sum, but with one critical rule: **every element is uniquely represented**. This means if you write a vector $\alpha \in W_1 \oplus W_2$ as

$$\alpha = \alpha_1 + \beta_1$$

, there is absolutely no other pair of vectors from W_1 and W_2 that can sum to α.

5.5.1 The Direct Sum Characterization Theorem

This theorem gives us an easy way to check if a regular sum is actually a direct sum.
Theorem: A vector space V is the direct sum of W_1 and W_2 (

$$V = W_1 \oplus W_2$$

) if and only if two conditions hold: 1.

$$V = W_1 + W_2$$

(It is a linear sum) 2.

$$W_1 \cap W_2 = \{\bar{0}\}$$

(Their intersection contains only the zero vector)
Proof Part 1: Condition is necessary Assume

$$V = W_1 \oplus W_2.$$

By definition, every element is a sum, so condition 1 (

$$V = W_1 + W_2$$

) holds automatically. To prove condition 2, suppose for a contradiction that the intersection contains a non-zero vector $\alpha \neq \bar{0}$. Since α is in both W_1 and W_2, we can write α in two ways: * First way: $\alpha = \alpha + \bar{0}$ (where $\alpha \in W_1$ and $\bar{0} \in W_2$) * Second way: $\alpha = \bar{0} + \alpha$ (where $\bar{0} \in W_1$ and $\alpha \in W_2$)

This gives us two different representations for the same vector α. But we assumed V is a direct sum, which requires uniqueness. This is a contradiction. Thus, our assumption was wrong, and $W_1 \cap W_2$ can only contain the zero vector.

Proof Part 2: Condition is sufficient Assume

$$V = W_1 + W_2$$

and

$$W_1 \cap W_2 = \{\bar{0}\}$$

. We must prove the representation is unique. Let us suppose a vector α has two representations:

$$\alpha = \alpha_1 + \beta_1$$

$$\alpha = \alpha_2 + \beta_2$$

where $\alpha_1, \alpha_2 \in W_1$ and $\beta_1, \beta_2 \in W_2$.

Equate them:

$$\alpha_1 + \beta_1 = \alpha_2 + \beta_2$$

Rearrange to put W_1 elements on the left and W_2 elements on the right:

$$\alpha_1 - \alpha_2 = \beta_2 - \beta_1$$

Since W_1 is a subspace, the left side $(\alpha_1 - \alpha_2)$ is in W_1. Since W_2 is a subspace, the right side $(\beta_2 - \beta_1)$ is in W_2. Because the two sides are equal, this element must belong to both W_1 and W_2:

$$\alpha_1 - \alpha_2 \in W_1 \cap W_2$$

But we assumed the intersection only contains the zero vector. So:

$$\alpha_1 - \alpha_2 = \bar{0} \implies \alpha_1 = \alpha_2$$

$$\beta_2 - \beta_1 = \bar{0} \implies \beta_1 = \beta_2$$

Since the elements are exactly the same, the representation is unique. Therefore,

$$V = W_1 \oplus W_2.$$

5.6 Key Takeaways

- A **linear combination** scales vectors and adds them together.
- The **linear span** is the set of all possible linear combinations of a set.
- The **linear sum** $W_1 + W_2$ is always a subspace.
- A **direct sum** $W_1 \oplus W_2$ is a sum where every element is uniquely formed.
- A sum is a direct sum if and only if the subspaces share nothing but the zero vector (

$$W_1 \cap W_2 = \{\bar{0}\}$$

).

5.7 What Comes Next

Next, we will discuss linear independence and dependence, which helps us understand when vectors are truly unique or if they are just linear combinations of each other.

6 Lecture 06: Vector Space - Linear Independence and Dependence of Vectors

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 06 of 27 **Video**: https://www.youtube.com/watch?v=9WjuLvPT_2A

6.1 Prerequisites

We know how to form a linear combination of vectors:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n$$

. Today, we will ask a specific question: what happens if we set this combination equal to the zero vector? The answer tells us whether the vectors are independent or dependent.

6.2 Definitions

Let $V(F)$ be a vector space. Let

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

be a finite set of vectors in $V(F)$. We form their linear combination and set it equal to the zero vector:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n = \bar{0}$$

where $a_1, a_2, \ldots, a_n \in F$ are scalars.

6.2.1 Linearly Independent (L.I.)

If the **only** way to satisfy this equation is for all the scalars to be zero:

$$a_1 = 0, a_2 = 0, \ldots, a_n = 0$$

then the vectors

$$\alpha_1, \alpha_2, \ldots, \alpha_n$$

are said to be **Linearly Independent**. This means none of the vectors can be built using the others. They are completely unique directions in the space.

6.2.2 Linearly Dependent (L.D.)

If there is at least one scalar $a_i \neq 0$ that satisfies the equation:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n = \bar{0}$$

then the vectors are called **Linearly Dependent**. This means at least one vector is redundant. It can be formed by mixing the other vectors together.

6.3 Theorem: Dependence Means Scalar Multiples

We can prove exactly what it means to be linearly dependent with a simple theorem.

Theorem: If two vectors are linearly dependent, then one can be represented as a scalar multiple of the other.

Proof: Let $\alpha, \beta \in V(F)$ be two linearly dependent vectors. By the definition of linear dependence, there exist scalars $a, b \in F$, which are **not both zero**, such that:

$$a\alpha + b\beta = \bar{0}$$

Since at least one scalar is not zero, let us check the cases.
Case 1: If $a \neq 0$ We can divide by a because it is non-zero. Move the β term to the right side:

$$a\alpha = -b\beta$$

Divide by a:

$$\alpha = -\left(\frac{b}{a}\right)\beta$$

Here, α is represented as a scalar multiple of β.
Case 2: If $b \neq 0$ We can divide by b. Move the α term to the right side:

$$b\beta = -a\alpha$$

Divide by b:

$$\beta = -\left(\frac{a}{b}\right)\alpha$$

Here, β is represented as a scalar multiple of α.
In both cases, one vector is just a scaled version of the other. The theorem is proven.

6.4 How to Test for Independence (Numerical Methods)

When you are given actual vectors with numbers, you must determine if they are L.I. or L.D. You can do this by solving equations directly, or by using a matrix determinant.

6.4.1 Method 1: Solving Equations

Question: Check if the vectors are L.I. or L.D. in $V_3(\mathbb{R})$:

$$\alpha_1 = (1, 2, 3), \quad \alpha_2 = (1, 0, 0), \quad \alpha_3 = (0, 1, 0), \quad \alpha_4 = (0, 0, 1)$$

Solution: Set up the linear combination with scalars

$$a_1, a_2, a_3, a_4 \in \mathbb{R}:$$

$$a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3 + a_4\alpha_4 = \bar{0}$$

Substitute the vectors:

$$a_1(1, 2, 3) + a_2(1, 0, 0) + a_3(0, 1, 0) + a_4(0, 0, 1) = (0, 0, 0)$$

Multiply the scalars and add the components:

$$(a_1 + a_2, 2a_1 + a_3, 3a_1 + a_4) = (0, 0, 0)$$

Compare components to get three equations:

$$a_1 + a_2 = 0 \implies a_2 = -a_1$$

$$2a_1 + a_3 = 0 \implies a_3 = -2a_1$$

$$3a_1 + a_4 = 0 \implies a_4 = -3a_1$$

Can we find a non-zero solution? Let us assume

$$a_1 = 1.$$

Then we get

$$a_2 = -1,$$

$$a_3 = -2$$

, and

$$a_4 = -3.$$

Since we found a valid solution where the scalars are not zero, the vectors are **Linearly Dependent**.

6.4.2 Method 2: The Determinant and Rank Method

When you have the same number of vectors as dimensions (e.g., three vectors in 3D space), you can use a matrix determinant.

Question: Check if the vectors are L.I. or L.D. in $V_3(\mathbb{R})$:

$$\alpha_1 = (2, 3, -1), \quad \alpha_2 = (-1, 4, -2), \quad \alpha_3 = (1, 18, -4)$$

Solution: Set up the combination:

$$a_1(2, 3, -1) + a_2(-1, 4, -2) + a_3(1, 18, -4) = (0, 0, 0)$$

This gives us a system of homogeneous equations:

$$2a_1 - a_2 + a_3 = 0$$

$$3a_1 + 4a_2 + 18a_3 = 0$$

$$-a_1 - 2a_2 - 4a_3 = 0$$

We build the coefficient matrix A:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 4 & 18 \\ -1 & -2 & -4 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$|A| = 2(4(-4) - 18(-2)) - (-1)(3(-4) - 18(-1)) + 1(3(-2) - 4(-1))$$

$$|A| = 2(-16 + 36) + 1(-12 + 18) + 1(-6 + 4)$$

$$|A| = 2(20) + 1(6) + 1(-2) = 40 + 6 - 2 = 44$$

Because the determinant is not zero (

$$|A| = 44 \neq 0$$

), the rank of the matrix is 3. When the rank equals the number of variables, the only solution to the homogeneous system is the trivial solution (

$$a_1 = 0, a_2 = 0, a_3 = 0$$

). Therefore, the vectors are **Linearly Independent**.

- If

$$|A| \neq 0 \implies \text{Rank} = 3 \implies \text{Linearly Independent}$$

- If

$$|A| = 0 \implies \text{Rank} < 3 \implies \text{Linearly Dependent}$$

6.5 Key Takeaways

- **Linearly Independent (L.I.)**: The combination

$$a_i\alpha_i = \bar{0}$$

is only solved by all zeros. * **Linearly Dependent (L.D.)**: The combination can be solved with non-zero scalars. * L.D. means vectors are redundant and can be written as multiples or combinations of each other. * For square systems, check the determinant. Non-zero means independent; zero means dependent.

6.6 What Comes Next

We will soon use the concepts of linear independence and span to define the "Basis" of a vector space, which is the minimal set of vectors needed to build the entire space.

7 Lecture 07: Vector Space - Basis and Finite Dimensional Vector Space

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 07 of 27 **Video**: https://www.youtube.com/watch?v=pG-IAFOiOw4

7.1 Prerequisites

We know how to determine if a set of vectors is linearly independent (L.I.) or linearly dependent (L.D.). Today, we look at the properties of these sets and how they lead us to the concept of a basis.

7.2 Properties of Linearly Independent and Dependent Sets

Before defining a basis, let us establish four critical properties of vector sets:

1. **Subsets of L.I. Sets:** If a set of vectors S is linearly independent, then any subset S' taken from S is also linearly independent.
2. **Supersets of L.D. Sets:** If a set of vectors S is linearly dependent, then any superset S_1 (where $S \subseteq S_1$) is also linearly dependent. Adding more vectors to a dependent set cannot make it independent.
3. **Redundancy in L.D. Sets:** In any linearly dependent set of vectors

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

, there is always at least one vector α_i that can be written as a linear combination of the vectors preceding it. 4. **Breaking L.I. Sets:** If you take a linearly independent set S and add a new vector α_m that is already in the linear span of S, the new, larger set becomes linearly dependent.

7.3 Basis of a Vector Space

We know that the linear span $L(S)$ of a set S is the collection of all vectors you can build using S. If S can build the entire vector space $V(F)$ such that $L(S) = V(F)$, we say that S **spans** or **generates** $V(F)$.

A **Basis** is simply the most efficient, minimal set of vectors needed to build the entire space.

Let $V(F)$ be a vector space over the field F. A subset S of $V(F)$ is called a **basis** of $V(F)$ if it satisfies two conditions: 1. S **is a linearly independent set.** (No vector is redundant; there is no wasted effort). 2. $L(S) = V(F)$**.** (Every element of $V(F)$ can be represented as a linear combination of a finite number of elements of S).

7.3.1 Finite Dimensional Vector Space

A vector space $V(F)$ is called a **finite dimensional vector space** (or finitely generated vector space) if there exists a finite subset S such that $L(S) = V(F)$. If the set that generates the whole space has a finite number of elements, the space itself is finite dimensional.

Because a basis generates the space ($L(S) = V(F)$) and is linearly independent, any finite dimensional vector space will have a basis with a finite number of elements.

7.4 Dimension of a Vector Space

There are two fundamental theorems regarding the dimension of a space: 1. Every finite dimensional vector space has at least one basis. 2. While a vector space can have many different bases, **every basis will have exactly the same number of elements**.

Because this number is strictly fixed, we use it to define the dimension of the space.

Definition: The **dimension** of a finite dimensional vector space $V(F)$ is the exact number of elements in its basis. It is denoted by $\dim(V(F))$ or simply $\dim V$.

7.4.1 Examples of Standard Dimensions

- **1D Space:** For the real line $\mathbb{R}$, $\dim(\mathbb{R}) = 1$.
- **2D Space:** For the coordinate plane $\mathbb{R}^2$, the standard basis is $\{(1, 0), (0, 1)\}$. Thus,
$$\dim(\mathbb{R}^2) = 2.$$
- **3D Space:** For $\mathbb{R}^3$, the standard basis is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Thus,
$$\dim(\mathbb{R}^3) = 3.$$
- **N-Dimensional Space:** For $\mathbb{R}^n$ (the space of n-tuples), the standard basis has n elements. Thus,
$$\dim(\mathbb{R}^n) = n.$$

7.5 Numerical Examples: Finding a Basis

How do we prove a set of vectors is a basis, or find the basis of a generated subspace? We use matrices.

7.5.1 Example 1: Verifying a Basis

Question: Show that the set $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis for $\mathbb{R}^3(\mathbb{R})$.

Solution: We must check two things: Are there enough vectors to span the space, and are they linearly independent? Since
$$\dim(\mathbb{R}^3) = 3$$
, any basis must have exactly 3 vectors. Our set S has 3 vectors. So, we only need to prove they are linearly independent.

We form the coefficient matrix A using the vectors as rows:
$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$
Calculate the determinant $|A|$:
$$|A| = 1(1 \cdot 1 - 0 \cdot 1) - 0 + 0 = 1$$
Since
$$|A| = 1 \neq 0$$
, the vectors are linearly independent. Because we have 3 linearly independent vectors in a 3-dimensional space, they automatically span the space. Thus, S is a basis for $\mathbb{R}^3(\mathbb{R})$.

7.5.2 Example 2: Expressing a Vector in a Given Basis

Question: Show that the set $S = \{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ is a basis for $\mathbb{R}^3$, and express the standard basis vector $e_1 = (1, 0, 0)$ as a linear combination of these basis vectors.

Solution: If we can uniquely express e_1 as a linear combination of the vectors in S, it implies they span the space and are linearly independent. Let $a_1, a_2, a_3 \in \mathbb{R}$ be scalars such that:
$$e_1 = a_1(1, 2, 1) + a_2(2, 1, 0) + a_3(1, -1, 2)$$
Substitute $e_1 = (1, 0, 0)$:
$$(1, 0, 0) = (a_1 + 2a_2 + a_3, 2a_1 + a_2 - a_3, a_1 + 2a_3)$$
Comparing components gives a system of equations:
$$a_1 + 2a_2 + a_3 = 1$$
$$2a_1 + a_2 - a_3 = 0$$
$$a_1 + 2a_3 = 0$$

Solving this system yields:

$$a_1 = -\frac{2}{9}, \quad a_2 = \frac{5}{9}, \quad a_3 = \frac{1}{9}$$

Since we found a unique solution, the set forms a basis. The linear combination is:

$$e_1 = -\frac{2}{9}(1, 2, 1) + \frac{5}{9}(2, 1, 0) + \frac{1}{9}(1, -1, 2)$$

7.5.3 Example 3: Basis of Complex Numbers over Reals

Question: Show that the set $S = \{a + ib, c + id\}$ is a basis of $\mathbb{C}(\mathbb{R})$ if and only if $ad - bc \neq 0$.

Solution: To form a basis, the vectors must be linearly independent. Let $x_1, x_2 \in \mathbb{R}$ be scalars such that their linear combination is the zero vector:

$$x_1(a + ib) + x_2(c + id) = 0 + i0$$

Group the real and imaginary parts:

$$(ax_1 + cx_2) + i(bx_1 + dx_2) = 0 + i0$$

Equating the real and imaginary parts to zero gives two equations:

$$ax_1 + cx_2 = 0$$

$$bx_1 + dx_2 = 0$$

We can represent this homogeneous system with a coefficient matrix A:

$$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

The determinant of A is:

$$|A| = ad - bc$$

For the vectors to be linearly independent, the system must only have the trivial solution ($x_1 = 0, x_2 = 0$). This happens if and only if the determinant is non-zero. Therefore, the vectors are linearly independent (and hence form a basis for the 2-dimensional space $\mathbb{C}(\mathbb{R})$) if and only if:

$$|A| = ad - bc \neq 0$$

7.5.4 Example 4: Finding the Basis of a Subspace

Question: Find a basis and the dimension of the subspace W of $\mathbb{R}^4(\mathbb{R})$ generated by:

$$\alpha_1 = (1, -2, 5, -3), \quad \alpha_2 = (2, 3, 1, -4), \quad \alpha_3 = (3, 8, -3, 5)$$

Solution: These three vectors generate W. But to find the basis, we must remove any redundant vectors so only the linearly independent ones remain. We do this by reducing their matrix to Echelon form.

Construct matrix A with the vectors as rows:

$$A = \begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & 5 \end{bmatrix}$$

Apply elementary row operations to create zeros below the leading diagonal (Echelon form). 1. $R_2 \to R_2 - 2R_1$ 2. $R_3 \to R_3 - 3R_1$

This simplifies to:

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix}$$

Now, clear the 14 in the third row: 3. $R_3 \to R_3 - 2R_2$

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The third row became completely zero, meaning vector α_3 was redundant (it was a combination of α_1 and α_2). The non-zero rows left in the Echelon form give us our linearly independent basis vectors.

$$\text{Basis}(W) = \{(1, -2, 5, -3), (0, 7, -9, 2)\}$$

Because there are 2 vectors in this basis, the dimension is:

$$\dim(W) = 2$$

7.6 Key Takeaways

- A **basis** is a set of linearly independent vectors that spans the entire vector space.
- The **dimension** of a space is the exact number of elements in its basis.
- To check if vectors form a basis, place them in a matrix and check the determinant. If $|A| \neq 0$, they are a basis.
- To find the basis of a spanned subspace, place the generating vectors in a matrix, reduce to Echelon form, and the remaining non-zero rows form the basis.

7.7 What Comes Next

We now understand how the dimension of a single subspace is calculated. Next, we will learn how to calculate the dimension when we add two subspaces together using the Dimension Sum Theorem.

8 Lecture 08: Vector Space - Dimension of Sum of Subspaces

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 08 of 27 **Video**: https://www.youtube.com/watch?v=itCTFw0d4hs

8.1 Prerequisites

We know that the dimension of a vector space is the exact number of vectors in its basis. We also know how to add two vector subspaces together using the linear sum $W_1 + W_2$, and the direct sum $W_1 \oplus W_2$.

Today, we look at how to calculate the dimension of the resulting combined subspace.

8.2 Dimension of the Sum of Subspaces

When we take the linear sum of two subspaces W_1 and W_2, the dimension of the resulting new subspace is related to the individual dimensions of W_1 and W_2. However, simply adding their dimensions is not enough, because they might share some vectors. We must subtract the overlap (their intersection).

Theorem: Let W_1 and W_2 be two subspaces of a finite dimensional vector space $V(F)$. The dimension of their linear sum is given by:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

This formula works exactly like the principle of inclusion-exclusion in set theory (

$$|A \cup B| = |A| + |B| - |A \cap B|$$

). We add the dimensions of the two individual subspaces, but since the intersection $W_1 \cap W_2$ was counted twice (once in W_1 and once in W_2), we subtract its dimension to get the correct total.

8.3 Special Case: Direct Sum

What happens if the sum is a direct sum?

We know that if V is the direct sum of two subspaces,

$$V = W_1 \oplus W_2$$

, it means that every element has a unique representation. By the direct sum characterization theorem, this uniqueness requires that their intersection contains only the zero vector:

$$W_1 \cap W_2 = \{\bar{0}\}$$

Since the set containing only the zero vector has a dimension of exactly zero:

$$\dim(W_1 \cap W_2) = 0$$

We substitute this zero back into our main formula:

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2) - 0$$

This gives us a very clean and simple result. For a direct sum, the dimension of the combined space is simply the sum of the individual dimensions.

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2)$$

8.4 Numerical Examples

Let us verify this formula using practical numerical examples.

8.4.1 Example 1: Sum in 3D Space

Consider the three-dimensional vector space $V_3(\mathbb{R})$. Let us take two subspaces:

$$W_1 = \{(a, 0, 0) \mid a \in \mathbb{R}\}$$

$$W_2 = \{(0, a, b) \mid a, b \in \mathbb{R}\}$$

Step 1: Find individual dimensions * W_1 has only one free variable (a), so it is a one-dimensional subspace:

$$\dim(W_1) = 1.$$

- W_2 has two free variables (a, b), so it is a two-dimensional subspace:

$$\dim(W_2) = 2.$$

Step 2: Find the intersection What points do W_1 and W_2 have in common? W_1 requires the last two coordinates to be zero. W_2 requires the first coordinate to be zero. To be in both sets, all three coordinates must be zero.

$$W_1 \cap W_2 = \{(0, 0, 0)\}$$

Since the intersection contains only the zero vector, its dimension is 0:

$$\dim(W_1 \cap W_2) = 0$$

Step 3: Calculate the dimension of the sum Using our formula:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

$$\dim(W_1 + W_2) = 1 + 2 - 0 = 3$$

The sum creates a 3-dimensional subspace, which covers the entire space $V_3(\mathbb{R})$. Because the intersection was zero, this was technically a direct sum.

8.4.2 Example 2: Sum with Overlap in 4D Space

Now consider the four-dimensional vector space $V_4(\mathbb{R})$. Let us take two overlapping subspaces:

$$W_1 = \{(a, b, 0, 0) \mid a, b \in \mathbb{R}\}$$

$$W_2 = \{(0, a, b, c) \mid a, b, c \in \mathbb{R}\}$$

Step 1: Find individual dimensions * W_1 has two free variables, so

$$\dim(W_1) = 2.$$

- W_2 has three free variables, so

$$\dim(W_2) = 3.$$

Step 2: Find the intersection What points do they share? W_1 says the 3rd and 4th coordinates must be zero. W_2 says the 1st coordinate must be zero. The 2nd coordinate is free to be anything in both sets. Thus, the intersection is:

$$W_1 \cap W_2 = \{(0, a, 0, 0) \mid a \in \mathbb{R}\}$$

Because this intersection has one free variable, it is a one-dimensional subspace.

$$\dim(W_1 \cap W_2) = 1$$

Step 3: Calculate the dimension of the sum

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

$$\dim(W_1 + W_2) = 2 + 3 - 1 = 4$$

The linear sum results in a 4-dimensional subspace, which is the entire space $V_4(\mathbb{R})$.

8.5 Key Takeaways

- The dimension of the linear sum of two subspaces is:
$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$
- If the sum is a **direct sum**, the intersection is zero, so the dimension is just the sum of the individual dimensions.
- Always find the exact structure of the intersection to determine its dimension before calculating the sum.

8.6 What Comes Next

We will prove several formal theorems regarding the linear span. These proofs will solidify our understanding of how vectors generate spaces before we move on to advanced theorems connecting independence and basis extension.

9 Lecture 09: Vector Space - Theorems on Linear Span

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 09 of 27 **Video**: https://www.youtube.com/watch?v=iF7ya80h1Rs

9.1 Prerequisites

We know that the linear span $L(S)$ of a set S is the collection of all possible linear combinations of the vectors in S. We also proved earlier that the linear span of any set is always a vector subspace of $V(F)$.

Today, we will prove five critical theorems related to the behavior of the linear span. These theorems test your understanding of how sets, spans, and subspaces interact.

9.2 The Five Theorems of Linear Span

Let S and T be two subsets of a vector space $V(F)$. The following five properties always hold true:

1. If $S \subseteq L(T)$, then $L(S) \subseteq L(T)$.
2. If $S \subseteq T$, then $L(S) \subseteq L(T)$.
3. S is a subspace of $V(F)$ if and only if $L(S) = S$.
4. $L(S \cup T) = L(S) + L(T)$.
5. $L(L(S)) = L(S)$.

We will prove each of these step by step. The key to proving these is to rely on two facts: * Any set is a subset of its own linear span ($S \subseteq L(S)$). * A linear span is always a subspace, meaning it is closed under linear combinations.

9.2.1 Proof 1: If, then

We are given that every element of S is already in the linear span of T. We want to show that any linear combination built from S will also belong to the linear span of T.

Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}.$$

Since $S \subseteq L(T)$, we know that all these vectors belong to $L(T)$:

$$\alpha_1, \alpha_2, \dots, \alpha_n \in L(T)$$

Now, let x be an arbitrary element of $L(S)$. By definition, x is a linear combination of elements of S:

$$x = a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n \quad \text{(where } a_i \in F\text{)}$$

We already know that $\alpha_1, \dots, \alpha_n$ belong to $L(T)$. We also know that $L(T)$ is a subspace. Because it is a subspace, it is closed under vector addition and scalar multiplication. Therefore, any linear combination of elements inside $L(T)$ must also stay inside $L(T)$.

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n \in L(T)$$

Because this combination is equal to x, we have:

$$x \in L(T)$$

We picked x from $L(S)$ and showed it belongs to $L(T)$. Thus:

$$L(S) \subseteq L(T)$$

9.2.2 Proof 2: If, then

This proof is almost identical to the first one.

Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}.$$

Since $S \subseteq T$, we know:

$$\alpha_1, \alpha_2, \dots, \alpha_n \in T$$

Any set is automatically a subset of its own linear span ($T \subseteq L(T)$). Because these vectors are in T, they must also be in $L(T)$:

$$\alpha_1, \alpha_2, \dots, \alpha_n \in L(T)$$

Let x be an arbitrary element of $L(S)$. By definition:

$$x = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$$

Because all the α_i vectors belong to the subspace $L(T)$, their linear combination must also belong to $L(T)$.

$$x \in L(T)$$

Thus, $L(S) \subseteq L(T)$.

9.2.3 Proof 3: is a subspace if and only if

This is an "if and only if" proof, so we must prove both directions.

Part A: Assume $L(S) = S$ We know that the linear span $L(S)$ of any set is always a vector subspace. Since S is equal to $L(S)$, S must also be a vector subspace.

Part B: Assume S **is a subspace** We want to show $L(S) = S$. To show two sets are equal, we prove they contain each other.

1. **Show** $S \subseteq L(S)$**:** Any vector $\alpha \in S$ can be written as $1 \cdot \alpha$. This is a linear combination of elements in S. Thus, $\alpha \in L(S)$. So, $S \subseteq L(S)$.
2. **Show** $L(S) \subseteq S$**:** Let $x \in L(S)$. Then x is a linear combination:

$$x = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n \quad (\text{where } \alpha_i \in S)$$

`Because $S$ is given as a subspace, it is closed under linear combinations. Therefore, this sum must bel`

$$x \in S$$

`Thus, $L(S) \subseteq S$.`

Since they contain each other, $L(S) = S$.

9.2.4 Proof 4

We want to show that the span of the union of two sets is the same as the linear sum of their individual spans. We prove mutual containment.

Part A: Show $L(S \cup T) \subseteq L(S) + L(T)$ Let $x \in L(S \cup T)$. This means x is a linear combination of elements from both S and T.

$$x = \sum_{i=1}^{n} a_i\alpha_i + \sum_{j=1}^{m} b_j\beta_j \quad (\text{where } \alpha_i \in S, \beta_j \in T)$$

The first sum is built entirely from elements of S, so it belongs to $L(S)$. Let's call it y. The second sum is built entirely from elements of T, so it belongs to $L(T)$. Let's call it z.

$$x = y + z \quad (\text{where } y \in L(S), z \in L(T))$$

By the definition of the linear sum of subspaces, x belongs to $L(S) + L(T)$. Thus:

$$L(S \cup T) \subseteq L(S) + L(T)$$

Part B: Show $L(S) + L(T) \subseteq L(S \cup T)$ Let $x \in L(S) + L(T)$. This means x is the sum of an element from $L(S)$ and an element from $L(T)$:

$$x = y + z$$

Since $y \in L(S)$, y is a linear combination of elements of S:

$$y = \sum a_i \alpha_i.$$

Since $z \in L(T)$, z is a linear combination of elements of T:

$$z = \sum b_j \beta_j.$$

Therefore,

$$x = \sum a_i \alpha_i + \sum b_j \beta_j.$$

Since both S and T are subsets of $S \cup T$, all α_i and β_j vectors belong to $S \cup T$. Thus, x is a linear combination of elements in $S \cup T$.

$$x \in L(S \cup T)$$

Thus, $L(S) + L(T) \subseteq L(S \cup T)$. Both directions are proven.

9.2.5 Proof 5

This looks complicated, but it is the easiest one to prove if we use Theorem 3.

From Theorem 3, we proved that for any subspace W, $L(W) = W$. We already know that the linear span $L(S)$ is always a subspace of $V(F)$. So, substitute $W = L(S)$ into the rule:

$$L(L(S)) = L(S)$$

The theorem is instantly proven.

9.3 Key Takeaways

- The linear span is always a subspace, so any linear combination of its elements stays inside it.
- If S is already a subspace, its span is just itself: $L(S) = S$.
- The span of a union is the linear sum of the spans: $L(S \cup T) = L(S) + L(T)$.
- Taking the span of a span does nothing new: $L(L(S)) = L(S)$.

9.4 What Comes Next

10 Lecture 10: Vector Space - Theorems on Linear Independence and Dependence

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 10 of 27 **Video**: https://www.youtube.com/watch?v=wGiJkeuJDcM

10.1 Prerequisites

We know the basic definitions of linear independence and dependence. A set of vectors is **Linearly Independent (L.I.)** if the only way their linear combination equals the zero vector is when all scalars are zero. They are **Linearly Dependent (L.D.)** if you can make the combination equal to zero using at least one non-zero scalar.

Today, we will prove several theorems that describe the behavior of L.I. and L.D. sets.

10.2 Basic Properties

10.2.1 Property 1: The Zero Vector is Always Linearly Dependent

If a set contains only the zero vector, is it independent or dependent? Let us form a linear combination:

$$a \cdot \bar{0} = \bar{0}$$

We can choose any non-zero scalar for a, for example, $a = 2$.

$$2 \cdot \bar{0} = \bar{0}$$

Because we found a non-zero scalar that satisfies the equation, the zero vector is, by definition, **Linearly Dependent**.

10.2.2 Property 2: Any Set Containing the Zero Vector is Linearly Dependent

If a larger set of vectors contains the zero vector, the entire set becomes infected and is classified as linearly dependent.

Proof: Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$$

be a set of vectors. Without loss of generality, let the first vector be the zero vector:

$$\alpha_1 = \bar{0}.$$

We form the linear combination:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n = \bar{0}$$

Because

$$\alpha_1 = \bar{0}$$

, we can choose its scalar a_1 to be any non-zero number, say

$$a_1 = 1$$

. For all the other vectors, we can just choose their scalars to be 0.

$$1 \cdot \bar{0} + 0 \cdot \alpha_2 + \cdots + 0 \cdot \alpha_n = \bar{0}$$

Since not all scalars are zero (specifically,

$$a_1 = 1 \neq 0$$

), the set S is linearly dependent.

10.3 Subsets and Supersets

10.3.1 Theorem: Any Subset of a Linearly Independent Set is Linearly Independent

If a large group of vectors are all completely independent of each other, taking a smaller group from them will not suddenly make them dependent.

Proof: Let

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

be a linearly independent set. By definition, if:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n = \bar{0} \implies a_1 = a_2 = \cdots = a_n = 0$$

Let S_1 be a subset of S containing the first k vectors:

$$S_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_k\}$$

where $1 < k \leq n$. We want to prove S_1 is linearly independent. We set up its linear combination:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_k\alpha_k = \bar{0}$$

We can rewrite this combination to look like the full set S by adding the missing vectors and giving them scalar coefficients of exactly zero:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_k\alpha_k + 0 \cdot \alpha_{k+1} + \cdots + 0 \cdot \alpha_n = \bar{0}$$

This is now a linear combination of the full independent set S that equals zero. Because S is linearly independent, *every single scalar* in this equation must be zero. Therefore:

$$a_1 = 0, \quad a_2 = 0, \quad \ldots, \quad a_k = 0$$

Since the only solution for the subset's combination is all zeros, the subset S_1 is linearly independent.

10.3.2 Lemma: Any Superset of a Linearly Dependent Set is Linearly Dependent

If a set is already linearly dependent (meaning it has redundant vectors), adding more vectors to it will not magically fix the redundancy.

Proof: Let

$$S = \{\alpha_1, \ldots, \alpha_n\}$$

be linearly dependent. This means there exist scalars $a_1, \ldots, a_n$, not all zero, such that:

$$a_1\alpha_1 + \cdots + a_n\alpha_n = \bar{0}$$

Let us add a new vector α to create a superset

$$S_1 = \{\alpha_1, \ldots, \alpha_n, \alpha\}.$$

We can write a combination for S_1:

$$a_1\alpha_1 + \cdots + a_n\alpha_n + 0 \cdot \alpha = \bar{0}$$

Because the original scalars $a_1, \ldots, a_n$ were not all zero, the scalars in this new equation are also not all zero. Therefore, the superset S_1 is linearly dependent.

10.4 The Preceding Vector Theorem

This is the most important theorem regarding linear dependence. It states exactly *why* a set is dependent.

Theorem: A set of non-zero vectors

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

is linearly dependent **if and only if** some vector α_k $(2 \leq k \leq n)$ can be expressed as a linear combination of its preceding vectors (

$$\alpha_1, \alpha_2, \ldots, \alpha_{k-1}$$

).

Because this is an "if and only if" theorem, we prove both directions.

10.4.1 Part 1: Assume is Linearly Dependent

We must show one vector is a combination of the ones before it. Because S is linearly dependent, there exist scalars $a_1, \ldots, a_n$ (not all zero) such that:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n = \bar{0}$$

Since the vectors are non-zero, $\alpha_1 \neq \bar{0}$. Let k be the largest index such that its scalar $a_k \neq 0$. (This means for all vectors after k, their scalars are zero:

$$a_{k+1} = 0, \ldots, a_n = 0$$

). We also know $k > 1$. (If $k = 1$, then

$$a_1\alpha_1 = \bar{0}$$

. Since $\alpha_1 \neq \bar{0}$, a_1 must be 0, which contradicts our assumption that not all scalars are zero).
So our equation simplifies to:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_k\alpha_k = \bar{0} \quad \text{(where } a_k \neq 0\text{)}$$

Because $a_k \neq 0$, we can isolate α_k. First, move the other terms to the right:

$$a_k\alpha_k = -a_1\alpha_1 - a_2\alpha_2 - \cdots - a_{k-1}\alpha_{k-1}$$

Divide everything by a_k:

$$\alpha_k = \left(-\frac{a_1}{a_k}\right)\alpha_1 + \left(-\frac{a_2}{a_k}\right)\alpha_2 + \cdots + \left(-\frac{a_{k-1}}{a_k}\right)\alpha_{k-1}$$

Thus, we have successfully expressed α_k as a linear combination of its preceding vectors.

10.4.2 Part 2: Assume is a combination of preceding vectors

We must show the set S is linearly dependent. Assume:

$$\alpha_k = c_1\alpha_1 + c_2\alpha_2 + \cdots + c_{k-1}\alpha_{k-1}$$

Move α_k to the right side of the equation:

$$c_1\alpha_1 + c_2\alpha_2 + \cdots + c_{k-1}\alpha_{k-1} - 1 \cdot \alpha_k = \bar{0}$$

Look closely at this linear combination. The scalar coefficient for α_k is exactly -1. Since $-1 \neq 0$, the scalars in this combination are **not all zero**. Therefore, the subset

$$\{\alpha_1, \ldots, \alpha_k\}$$

is linearly dependent.
Since any superset of a linearly dependent set is also linearly dependent (as proven in our lemma), the full set

$$S = \{\alpha_1, \ldots, \alpha_n\}$$

is linearly dependent.

10.5 Key Takeaways

- The zero vector destroys independence. Any set containing it is linearly dependent.
- **Subsets** of independent sets remain independent.
- **Supersets** of dependent sets remain dependent.
- A set is linearly dependent if and only if you can build one of its vectors using only the vectors that came before it.

10.6 What Comes Next

We will use these theorems to prove the Existence Theorem, which guarantees that every finitely generated vector space has a basis, and then we will look at the Extension Theorem.

11 Lecture 11: Vector Space - Existence and Dimension Theorem

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 11 of 27 **Video**: https://www.youtube.com/watch?v=cwIIXBSEFX4

11.1 Prerequisites

We know a vector space is **finite-dimensional** if it can be completely generated by a finite set of vectors. We also know a **basis** is a set of vectors that generates the space AND is linearly independent. Today, we prove two fundamental theorems: 1. **Existence Theorem**: Does every finite-dimensional space actually have a basis? 2. **Dimension Theorem**: If a space has multiple bases, do they all have the exact same number of elements?

11.2 The Existence Theorem

Statement: There exists a basis for each finite-dimensional vector space.

Proof: Let $V(F)$ be a finite-dimensional vector space. By the definition of "finite-dimensional," there must exist a finite subset

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

that generates the entire space. This means:

$$L(S) = V$$

We assume no vector in S is the zero vector. (If there is a zero vector, we can throw it out because it contributes nothing to the linear span).

Now, there are two possibilities for this generating set S: **Case 1: S is Linearly Independent** If S is already linearly independent, then since it also generates V, it perfectly matches the definition of a basis. The theorem is proven immediately.

Case 2: S is Linearly Dependent If S is linearly dependent, we can use our previous theorem: there exists some vector $\alpha_i \in S$ that can be expressed as a linear combination of its preceding vectors.

$$\alpha_i = a_1\alpha_1 + a_2\alpha_2 + \cdots + a_{i-1}\alpha_{i-1} \quad \text{— (1)}$$

Because α_i is redundant (it is built entirely from the other vectors), we can omit it from the set S without losing any generating power. Let us create a new set S_1 by removing α_i:

$$S_1 = \{\alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_n\}$$

Does S_1 still generate V? Yes. Let us prove it. Pick any vector $\alpha \in V$. Since S generates V, we know:

$$\alpha = c_1\alpha_1 + \cdots + c_{i-1}\alpha_{i-1} + c_i\alpha_i + c_{i+1}\alpha_{i+1} + \cdots + c_n\alpha_n$$

Substitute the value of α_i from equation (1) into this combination:

$$\alpha = c_1\alpha_1 + \cdots + c_i(a_1\alpha_1 + \cdots + a_{i-1}\alpha_{i-1}) + \cdots + c_n\alpha_n$$

Group the terms together for each α:

$$\alpha = (c_1 + c_ia_1)\alpha_1 + \cdots + (c_{i-1} + c_ia_{i-1})\alpha_{i-1} + c_{i+1}\alpha_{i+1} + \cdots + c_n\alpha_n$$

Look at this new combination! The vector α is now written entirely using elements of S_1. Therefore, S_1 generates V. So,

$$L(S_1) = V.$$

If S_1 is linearly independent, it is our basis. If S_1 is still linearly dependent, we repeat the process. We find another redundant vector, remove it to make S_2, and S_2 will still generate V. Because our starting set S was finite (it had n elements), we cannot remove vectors forever. The process must stop. The worst-case scenario is that we are left with only one non-zero vector. A single non-zero vector is always linearly independent. Thus, we are guaranteed to eventually reach a subset that is linearly independent and generates V. This subset is a basis. Therefore, a basis always exists.

11.3 The Dimension Theorem

A vector space can have many different bases. But do they all share the same size?

Statement: If $V(F)$ is a finite-dimensional vector space, then any two bases of V have the same number of elements.

Proof: Let $V(F)$ be a finite-dimensional vector space. Let us pick two different bases for it:

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\} \quad (n \text{ elements})$$
$$S' = \{\beta_1, \beta_2, \ldots, \beta_m\} \quad (m \text{ elements})$$

We must prove that $m = n$.

Step 1: Focus on Basis S Because S is a basis, it generates the whole space ($L(S) = V$). Take the very first vector from the other basis, β_1. Because β_1 is in the vector space, it can be written as a linear combination of the elements in S:

$$\beta_1 = a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n$$

Step 2: Construct a new set S_1 Let us add β_1 to the beginning of set S:

$$S_1 = \{\beta_1, \alpha_1, \alpha_2, \ldots, \alpha_n\}$$

This new set S_1 has $n + 1$ elements. Because we know β_1 is a combination of the alphas, this set is linearly dependent.

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n - 1 \cdot \beta_1 = \bar{0}$$

Because the set is linearly dependent, some vector is a combination of its preceding vectors. This vector cannot be β_1 (since $\beta_1 \neq \bar{0}$). It must be one of the alphas, say α_i.

Step 3: Omit α_i Since α_i is redundant, we can throw it away.

$$S_2 = \{\beta_1, \alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_n\}$$

We replaced one α with one β. And just like in the Existence Theorem, this new set S_2 still generates V.

Step 4: Repeat the process Now take the second vector β_2. Since S_2 generates V, β_2 is a combination of elements in S_2. Add β_2 to the front to make S_3:

$$S_3 = \{\beta_2, \beta_1, \alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_n\}$$

This set is linearly dependent. The redundant vector cannot be β_2. It cannot be β_1 either, because the betas come from S', which is a basis, meaning the betas are strictly independent of each other. So the redundant vector MUST be another alpha, say α_j. We throw α_j away.

Step 5: The Logical Conclusion We continue this replacement process: Add a β vector, throw away a redundant α vector. What if the alphas run out before the betas? This would mean $n < m$. After all alphas are gone, we would be left with a set of betas:

$$\{\beta_n, \beta_{n-1}, \ldots, \beta_1\}$$

that generates the entire space V. But if $n < m$, this set is just a *part* of the full basis S'. A proper subset of a basis can never generate the entire space (because the remaining betas like β_m would have to be combinations of this subset, which breaks their independence). This creates a massive contradiction. Therefore, the alphas CANNOT run out before the betas. This strictly implies:

$$n \geq m \quad \text{— (2)}$$

Now, swap the roles of the two bases. Start with S' as the generating base, and insert alphas one by one. Using the exact same logic, the betas cannot run out before the alphas. This strictly implies:

$$m \geq n \quad \text{— (3)}$$

From equations (2) and (3), there is only one mathematical truth:

$$m = n$$

The theorem is proven. Any two bases must have exactly the same number of elements.

11.4 Key Takeaways

- **Existence Theorem**: Every finite-dimensional vector space has a basis. We find it by taking a generating set and throwing away redundant vectors until it becomes linearly independent.
- **Dimension Theorem**: Every basis of a specific vector space has the exact same number of elements. This fixed number is what we call the **dimension** of the space.
- The replacement technique (adding a vector, finding the set becomes dependent, and removing a different vector) is a core strategy in linear algebra proofs.

11.5 What Comes Next

We will look at the Extension Theorem, which tells us how to start with a linearly independent set and build it *up* into a full basis.

12 Lecture 12: Vector Space - Extension Theorem and Complementary Spaces

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 12 of 27 **Video**: https://www.youtube.com/watch?v=WBvroWQecEI

12.1 Prerequisites

We know that every finite-dimensional vector space has a basis (Existence Theorem) and that every basis of a specific space has exactly the same number of elements (Dimension Theorem). Today, we look at how to build a basis from scratch if we already have a few linearly independent vectors.

12.2 The Extension Theorem

What happens if you have a linearly independent set, but it is too small to span the entire space? Can you force it to become a basis?

Theorem: Every linearly independent subset of a finitely generated vector space $V(F)$ is either already a basis of V, or can be extended to form a basis of V.

Proof: Let $\dim(V) = n$. This means any basis of V must have exactly n elements. Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$$

be a linearly independent subset of V. Because it is linearly independent, $m \le n$. Let

$$S' = \{\beta_1, \beta_2, \dots, \beta_n\}$$

be a known basis of V.

We will merge these two sets by putting our independent alphas at the front:

$$S_1 = \{\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n\}$$

Because S' is a basis, it generates V. Since S' is completely inside S_1, S_1 must also generate V. But S_1 has $m + n$ elements, which is more than n, so it is definitely linearly dependent.

Since S_1 is linearly dependent, some vector inside it is a linear combination of its preceding vectors. Which vector is redundant? It **cannot** be one of the alphas, because the alphas form a linearly independent set on their own. Therefore, the redundant vector must be one of the betas. Let us call it β_i.

We remove β_i to form S_2:

$$S_2 = \{\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_{i-1}, \beta_{i+1}, \dots, \beta_n\}$$

Because the vector we removed was a linear combination of the others, S_2 still generates V. If S_2 is linearly independent, it is our basis. If not, we repeat the process. We find the next redundant beta vector and remove it.

We repeat this until the set becomes linearly independent. Because the final set must generate V and be linearly independent, it must contain exactly n elements (by the Dimension Theorem). Since we never remove any alphas, our final basis will contain all m alphas and exactly $n - m$ betas. Thus, our original linearly independent set S has been successfully extended into a basis.

12.3 The Maximum Size of Independent Sets

If you know the dimension of a space, you immediately know the absolute limit on how many independent vectors you can have.

Theorem: If $V(F)$ is a finitely generated vector space of dimension n, then any set containing $n + 1$ or more vectors is linearly dependent.

Proof (by contradiction): Assume we have a set S with $n+1$ vectors that is linearly independent. By the Extension Theorem, we can extend S to form a basis of V. Because S already has $n+1$ vectors, the resulting basis must have *at least* $n+1$ vectors. But the Dimension Theorem states that every basis of V must have exactly n vectors. A basis cannot have n vectors and $n+1$ vectors at the same time. This is a contradiction. Therefore, our assumption was wrong. Any set with $n+1$ or more vectors must be linearly dependent.

12.4 Shortcuts for Finding a Basis

If you know $\dim(V) = n$, you do not always need to check both conditions (independence and spanning) to prove a set of n vectors is a basis. You only need to check one!

12.4.1 Shortcut 1: Checking Independence Only

Theorem: If $\dim(V) = n$, then any set of exactly n linearly independent vectors in V forms a basis of V. **Proof:** Let S have n independent vectors. If S does not span V, we could use the Extension Theorem to add vectors to it to make it a basis. But adding vectors would make the basis have more than n elements, violating the Dimension Theorem. Thus, S must already span V. It is a basis.

12.4.2 Shortcut 2: Checking Spanning Only

Theorem: If $\dim(V) = n$, then any set of exactly n vectors that generates V is a basis of V. **Proof:** Let S have n generating vectors. If S were linearly dependent, we could remove redundant vectors and still generate V, leaving us with a generating set of fewer than n vectors. We could then shrink it to a basis with fewer than n elements, violating the Dimension Theorem. Thus, S must be linearly independent. It is a basis.

12.5 Unique Representation Theorem

Theorem: Let

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

be a basis of $V(F)$. Then any vector $\alpha \in V$ can be **uniquely** expressed as a linear combination of the elements of S.

Proof: Suppose α can be expressed in two different ways:

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n$$

$$\alpha = b_1\alpha_1 + b_2\alpha_2 + \cdots + b_n\alpha_n$$

Subtract the two equations:

$$\bar{0} = (a_1 - b_1)\alpha_1 + (a_2 - b_2)\alpha_2 + \cdots + (a_n - b_n)\alpha_n$$

Because S is a basis, its vectors are linearly independent. The only way their combination equals zero is if all scalar coefficients are exactly zero.

$$a_1 - b_1 = 0 \implies a_1 = b_1$$

$$a_n - b_n = 0 \implies a_n = b_n$$

The coefficients are identical. The representation is unique.

12.6 Subspaces and Dimensions

12.6.1 Dimension of a Subspace

Theorem: If W is a subspace of a finite-dimensional vector space $V(F)$, then $\dim(W) \leq \dim(V)$. **Proof:** Let $\dim(V) = n$. Because W is inside V, any independent set in W is also an independent set in V. Since no independent set in V can have more than n vectors, the largest independent set we can find in W can have at most n vectors. Let this maximal independent set have m vectors ($m \leq n$). Because it is maximal, adding any other vector from W makes it dependent, meaning it spans W. It is a basis for W. Thus, $\dim(W) = m \leq n$.

12.6.2 Existence of a Complementary Subspace

Theorem: If W_1 is a subspace of $V(F)$ where $\dim(V) = n$, there exists another subspace W_2 such that the whole space is their direct sum:

$$V = W_1 \oplus W_2$$

. Also,

$$\dim(W_2) = n - \dim(W_1).$$

Proof: Let

$$\dim(W_1) = m$$

. Let

$$S_1 = \{\alpha_1, \dots, \alpha_m\}$$

be a basis of W_1. Since S_1 is linearly independent in V, we use the Extension Theorem to expand it to a full basis of V:

$$S = \{\alpha_1, \dots, \alpha_m, \alpha_{m+1}, \dots, \alpha_n\}$$

We group the newly added vectors to form our complementary subspace:

$$W_2 = L(\{\alpha_{m+1}, \dots, \alpha_n\})$$

Because S is a basis for V, every vector in V can be built uniquely by taking some combination of the first m vectors (which lives in W_1) and adding it to a combination of the last $n - m$ vectors (which lives in W_2). Since the representation is unique,

$$V = W_1 \oplus W_2.$$

The dimension of W_2 is exactly $n - m$, which is $n - \dim(W_1)$.

12.7 Key Takeaways

- You can always build a full basis starting from any independent set.
- A space of dimension n cannot hold more than n independent vectors.
- If you have exactly n vectors in an n-dimensional space, checking just independence OR just spanning is enough to prove they are a basis.
- Coordinates (coefficients) relative to a basis are always unique.
- Every subspace has a complementary subspace that "fills in the rest" of the main vector space via direct sum.

13 Lecture 13: Vector Space - Dimension of Sum of Subspaces (Proof)

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 13 of 27 **Video**: https://www.youtube.com/watch?v=bJcGmEkiyz8

13.1 Prerequisites

We know the statement of the theorem for the dimension of the linear sum of two subspaces from earlier lectures. Today, we provide the full, formal proof of this theorem, verifying it with basis extension, and applying it to numerical examples.

13.2 Theorem Statement

If W_1 and W_2 are two subspaces of a finite-dimensional vector space $V(F)$, then:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

13.3 The Proof

This proof relies on picking a basis for the intersection of the subspaces, and extending it in two different directions (one for W_1 and one for W_2). Finally, we combine these extensions to form a basis for the sum $W_1 + W_2$.

13.3.1 Step 1: Define the Basis for the Intersection

Because $W_1 \cap W_2$ is a subspace, it has a basis. Let the dimension of this intersection be k:

$$\dim(W_1 \cap W_2) = k$$

Let the basis for this intersection be S:

$$S = \{\gamma_1, \gamma_2, \ldots, \gamma_k\}$$

13.3.2 Step 2: Extend the Basis for and

The intersection $W_1 \cap W_2$ is fully contained within W_1. This means the basis S is a linearly independent subset of W_1. By the **Extension Theorem**, we can extend S to form a complete basis for W_1 by adding n vectors:

$$S_1 = \{\gamma_1, \ldots, \gamma_k, \alpha_1, \alpha_2, \ldots, \alpha_n\}$$

Thus,

$$\dim(W_1) = k + n.$$

Similarly, the intersection $W_1 \cap W_2$ is fully contained within W_2. We can extend S by adding m vectors to form a complete basis for W_2:

$$S_2 = \{\gamma_1, \ldots, \gamma_k, \beta_1, \beta_2, \ldots, \beta_m\}$$

Thus,

$$\dim(W_2) = k + m.$$

Let's look at what we are trying to prove. The right side of our theorem formula is:

$$\dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) = (k + n) + (k + m) - k = k + n + m$$

To prove the theorem, we must prove that the left side, $\dim(W_1 + W_2)$, is also equal to $k + n + m$.

13.3.3 Step 3: Propose a Basis for

We propose that the set containing all unique vectors from S_1 and S_2 forms the basis for $W_1 + W_2$:

$$S' = \{\gamma_1, \dots, \gamma_k, \alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m\}$$

This set has exactly $k + n + m$ elements. We must prove it is a basis by showing it is linearly independent and spans $W_1 + W_2$.

13.3.3.1 Part A: Prove is Linearly Independent Set a linear combination of all vectors in S' equal to zero:

$$c_1\gamma_1 + \dots + c_k\gamma_k + a_1\alpha_1 + \dots + a_n\alpha_n + b_1\beta_1 + \dots + b_m\beta_m = \bar{0} \quad \text{— (1)}$$

We need to prove that all scalars c_i, a_j, b_l are strictly zero.
Isolate the β terms on the left side:

$$b_1\beta_1 + \dots + b_m\beta_m = -(c_1\gamma_1 + \dots + c_k\gamma_k + a_1\alpha_1 + \dots + a_n\alpha_n) \quad \text{— (2)}$$

Look at equation (2). The right-hand side is built entirely from gammas and alphas, which are the basis vectors of W_1. Therefore, this vector lives inside W_1. But the left-hand side is exactly equal to it, so the left-hand side must also live inside W_1:

$$b_1\beta_1 + \dots + b_m\beta_m \in W_1$$

However, the left-hand side is built entirely from betas. Betas belong to W_2. So:

$$b_1\beta_1 + \dots + b_m\beta_m \in W_2$$

Because this vector is in both W_1 and W_2, it must be in their intersection:

$$b_1\beta_1 + \dots + b_m\beta_m \in W_1 \cap W_2$$

We already defined the basis of the intersection as

$$S = \{\gamma_1, \dots, \gamma_k\}$$

. Thus, this vector can be written as a combination of gammas:

$$b_1\beta_1 + \dots + b_m\beta_m = d_1\gamma_1 + \dots + d_k\gamma_k \quad \text{(for some scalars } d_i\text{)}$$

Bring everything to one side:

$$b_1\beta_1 + \dots + b_m\beta_m - d_1\gamma_1 - \dots - d_k\gamma_k = \bar{0}$$

This is a linear combination of vectors in S_2 (gammas and betas) equaling zero. Since S_2 is a basis for W_2, its elements are linearly independent. Therefore, every single scalar in this equation must be zero:

$$b_1 = 0, \dots, b_m = 0 \quad \text{and} \quad d_1 = 0, \dots, d_k = 0$$

Now substitute

$$b_1 = \dots = b_m = 0$$

back into our very first equation (1). It shrinks to:

$$c_1\gamma_1 + \dots + c_k\gamma_k + a_1\alpha_1 + \dots + a_n\alpha_n = \bar{0}$$

This is a linear combination of vectors in S_1 (gammas and alphas) equaling zero. Since S_1 is a basis for W_1, it is independent. Thus:

$$c_1 = 0, \dots, c_k = 0 \quad \text{and} \quad a_1 = 0, \dots, a_n = 0$$

We have proven that all scalars are zero. Therefore, S' is **Linearly Independent**.

13.3.3.2 Part B: Prove Spans

Let α be an arbitrary vector in $W_1 + W_2$. By definition, $\alpha = \alpha' + \beta'$, where $\alpha' \in W_1$ and $\beta' \in W_2$. Since S_1 is the basis for W_1, α' is a combination of gammas and alphas. Since S_2 is the basis for W_2, β' is a combination of gammas and betas.

When we add them together to get α, we get a giant combination of gammas, alphas, and betas. Since all of these vectors exist within S', we can build any vector $\alpha \in W_1 + W_2$ using S'. Thus, S' spans $W_1 + W_2$.

Because S' is linearly independent and spans $W_1 + W_2$, it is a basis for $W_1 + W_2$. Since S' has $k + n + m$ elements:

$$\dim(W_1 + W_2) = k + n + m = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

The theorem is proven.

13.4 Direct Sum Special Case

If the vector space is the direct sum $W_1 \oplus W_2$, the intersection only contains the zero vector, meaning its dimension is 0. The formula simplifies to:

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2)$$

13.5 Numerical Examples

13.5.1 Example 1: Sum in 3D Space

Let $V_3(\mathbb{R})$ be a 3D space with subspaces:

$$W_1 = \{(a, 0, 0) \mid a \in \mathbb{R}\} \implies \dim(W_1) = 1$$

$$W_2 = \{(0, b, c) \mid b, c \in \mathbb{R}\} \implies \dim(W_2) = 2$$

The intersection is elements where the 1st coordinate is 0 (from W_2) and the last two are 0 (from W_1).

$$W_1 \cap W_2 = \{(0, 0, 0)\} \implies \dim(W_1 \cap W_2) = 0$$

By the theorem:

$$\dim(W_1 + W_2) = 1 + 2 - 0 = 3$$

Since

$$\dim(W_1 + W_2) = 3 = \dim(V_3(\mathbb{R}))$$

, their sum spans the entire 3D space.

13.5.2 Example 2: Sum in 4D Space

Let $V_4(\mathbb{R})$ be a 4D space with subspaces:

$$W_1 = \{(a, b, 0, 0) \mid a, b \in \mathbb{R}\} \implies \dim(W_1) = 2$$

$$W_2 = \{(0, a, b, c) \mid a, b, c \in \mathbb{R}\} \implies \dim(W_2) = 3$$

The intersection requires the 1st coordinate to be zero (from W_2) and the last two to be zero (from W_1). Only the 2nd coordinate is free.

$$W_1 \cap W_2 = \{(0, x, 0, 0) \mid x \in \mathbb{R}\} \implies \dim(W_1 \cap W_2) = 1$$

By the theorem:

$$\dim(W_1 + W_2) = 2 + 3 - 1 = 4$$

Since their sum dimension matches the parent space dimension, their sum spans the entire 4D space.

13.6 What Comes Next

14 Lecture 14: Vector Space - Dimension of Quotient Space

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 14 of 27 **Video**: https://www.youtube.com/watch?v=HZxJtGqkJLM

14.1 Prerequisites

We are familiar with vector subspaces and their dimensions. Today, we will introduce a new mathematical structure built from a vector space and its subspace, known as a **Quotient Space**. We will define its operations and prove a fundamental theorem regarding its dimension.

14.2 Understanding Quotient Spaces

Let $V(F)$ be a vector space, and let W be a subspace of V. A quotient space is built by grouping vectors of V into "slices" or "cosets" based on the subspace W.

Definition: The quotient space V/W is the set of all cosets of W in V.

$$V/W = \{W + \alpha \mid \alpha \in V\}$$

Here, $W + \alpha$ is a coset formed by taking every vector in W and adding the specific vector α to it. You can think of it as shifting the entire subspace W by the vector α.

14.2.1 Operations in

To make V/W a vector space itself, we must define vector addition and scalar multiplication for these cosets.

1. **Vector Addition**: To add two cosets, you simply add their representative vectors.
$$(W + \alpha) + (W + \beta) = W + (\alpha + \beta)$$
2. **Scalar Multiplication**: To scale a coset, you scale its representative vector.
$$a(W + \alpha) = W + a\alpha$$

The Identity Element (The Zero Vector): In standard vector spaces, adding the zero vector changes nothing ($\alpha + \bar{0} = \alpha$). In the quotient space V/W, the "zero vector" is the subspace W itself (which is the same as $W + \bar{0}$). If you add W to any coset, nothing changes:

$$W + (W + \alpha) = (W + W) + \alpha = W + \alpha$$

Thus, W is the additive identity of the quotient space.

14.3 The Dimension of a Quotient Space

Since V/W is a vector space, it has a dimension. How is its dimension related to the dimensions of V and W?

Theorem: If W is a subspace of a finite-dimensional vector space $V(F)$, then:

$$\dim(V/W) = \dim(V) - \dim(W)$$

14.3.1 Proof:

Step 1: Setup Bases for W and V Let $\dim(W) = n$. This means W has a basis S with n elements:

$$S = \{\beta_1, \beta_2, \ldots, \beta_n\}$$

Because S is linearly independent in V, we can use the Extension Theorem to extend it to a full basis for V by adding m new vectors:

$$S' = \{\beta_1, \ldots, \beta_n, \alpha_1, \alpha_2, \ldots, \alpha_m\}$$

Because S' has $n+m$ elements, we know $\dim(V) = n+m$.
If we look at the formula we want to prove:

$$\dim(V) - \dim(W) = (n+m) - n = m$$

We need to prove that $\dim(V/W) = m$. To do this, we must find a basis for V/W that has exactly m elements.

Step 2: Propose a Basis for V/W We propose that the cosets formed by the newly added α vectors form the basis for V/W.

$$S_1 = \{W + \alpha_1, W + \alpha_2, \dots, W + \alpha_m\}$$

We must prove that S_1 is linearly independent and that it spans V/W.

Step 3: Prove S_1 is Linearly Independent Set a linear combination of the elements of S_1 to the zero element of the quotient space (which is W):

$$a_1(W + \alpha_1) + a_2(W + \alpha_2) + \cdots + a_m(W + \alpha_m) = W$$

Apply the operations of scalar multiplication and addition:

$$W + (a_1\alpha_1 + a_2\alpha_2 + \cdots + a_m\alpha_m) = W$$

We know from group theory that $H + x = H$ if and only if $x \in H$. Therefore, the vector inside the parenthesis must belong to W:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_m\alpha_m \in W$$

Since it belongs to W, it can be written as a linear combination of the basis vectors of W (the β vectors):

$$a_1\alpha_1 + \cdots + a_m\alpha_m = b_1\beta_1 + \cdots + b_n\beta_n \quad \text{(for some scalars } b_i\text{)}$$

Bring all terms to one side:

$$a_1\alpha_1 + \cdots + a_m\alpha_m - b_1\beta_1 - \cdots - b_n\beta_n = \bar{0}$$

This is a combination of vectors from S' equating to zero. Because S' is the basis of V, its vectors are linearly independent. Thus, every single scalar in this equation must be exactly zero:

$$a_1 = \cdots = a_m = 0 \quad \text{and} \quad b_1 = \cdots = b_n = 0$$

Because all

$$a_i = 0$$

, our set of cosets S_1 is linearly independent.

Step 4: Prove S_1 Spans V/W Take any arbitrary coset $W + \alpha \in V/W$. Because $\alpha \in V$, we can write α using the basis S':

$$\alpha = (a_1\alpha_1 + \cdots + a_m\alpha_m) + (b_1\beta_1 + \cdots + b_n\beta_n)$$

Let the second grouped part be

$$w = b_1\beta_1 + \cdots + b_n\beta_n$$

. Since this is built only from β vectors, $w \in W$. So,

$$\alpha = a_1\alpha_1 + \cdots + a_m\alpha_m + w.$$

Now substitute this into our coset:

$$W + \alpha = W + (a_1\alpha_1 + \cdots + a_m\alpha_m + w)$$

Because $w \in W$, adding it to W absorbs it ($W + w = W$):

$$W + \alpha = W + (a_1\alpha_1 + \cdots + a_m\alpha_m)$$

Using our quotient space operations, we can break this apart:

$$W + \alpha = a_1(W + \alpha_1) + a_2(W + \alpha_2) + \cdots + a_m(W + \alpha_m)$$

This proves that any coset $W + \alpha$ can be built as a linear combination of the elements in S_1. Thus, S_1 spans V/W.

Because S_1 is independent and spans V/W, it is a basis. Because S_1 has m elements:

$$\dim(V/W) = m = \dim(V) - \dim(W)$$

14.4 Numerical Application: Dimensions of Sums and Intersections

Let's use our skills to solve a matrix rank problem relating to subspace dimensions.

Question: Let W_1 and W_2 be two subspaces of $V_4(\mathbb{R})$. W_1 is generated by

$$S_1 = \{(1,1,0,-1),(1,2,3,0),(2,3,3,-1)\}$$

W_2 is generated by

$$S_2 = \{(1,2,2,-2),(2,3,2,-3),(1,3,4,-3)\}$$

Find $\dim(W_1 + W_2)$ and $\dim(W_1 \cap W_2)$.

Solution: The dimension of a subspace generated by a set of vectors is exactly equal to the rank of the matrix formed by those vectors.

1. **Find** $\dim(W_1)$: Place the vectors of S_1 into a matrix and reduce to Echelon form:

$$A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 3 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since there are 2 non-zero rows, $\text{Rank}(A) = 2$. Thus,

$$\dim(W_1) = 2.$$

2. **Find** $\dim(W_2)$: Place vectors of S_2 into a matrix:

$$B = \begin{bmatrix} 1 & 2 & 2 & -2 \\ 2 & 3 & 2 & -3 \\ 1 & 3 & 4 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 2 & -2 \\ 0 & -1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\text{Rank}(B) = 2$. Thus,

$$\dim(W_2) = 2.$$

3. **Find** $\dim(W_1 + W_2)$: The subspace $W_1 + W_2$ is generated by combining *all* the vectors from S_1 and S_2. We place all 6 vectors into a large 6×4 matrix. Upon reducing this matrix to Echelon form, we find the rank is 3. Thus,

$$\dim(W_1 + W_2) = 3.$$

4. **Find** $\dim(W_1 \cap W_2)$: We use the dimension sum theorem:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

$$3 = 2 + 2 - \dim(W_1 \cap W_2)$$

$$\dim(W_1 \cap W_2) = 4 - 3 = 1$$

14.5 What Comes Next

We have completed our extensive look into the structural dimensions of vector spaces, subspaces, and quotient spaces. Next, we will introduce **Linear Transformations**, which are functions that map vectors from one space to another while preserving their linear structure.

15 Lecture 15: Vector Space - Linear Transformation and its Properties

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 15 of 27 **Video**: https://www.youtube.com/watch?v=EL3fXu9FFII

15.1 Prerequisites

Until now, we have studied the internal structure of a single vector space (subspaces, span, basis, dimension). Now, we shift our focus to mappings *between* two different vector spaces.

Just as we study functions between sets in calculus, or homomorphisms between groups in group theory, we study **Linear Transformations** between vector spaces.

15.2 Definition of Linear Transformation

Let $U(F)$ and $V(F)$ be two vector spaces over the same underlying field F. A mapping $f : U \to V$ is called a **Linear Transformation** (or **Vector Space Homomorphism**) if it perfectly preserves the two fundamental operations of vector spaces: vector addition and scalar multiplication.

Specifically, it must satisfy two conditions: 1. **Preservation of Addition**: For any two vectors $x, y \in U$:

$$f(x + y) = f(x) + f(y)$$

`*(Adding vectors first then mapping them gives the same result as mapping them first then adding them i`

2. **Preservation of Scalar Multiplication**: For any scalar $a \in F$ and any vector $x \in U$:

$$f(ax) = af(x)$$

`*(Scaling a vector first then mapping it gives the same result as mapping it first then scaling it in t`

15.2.1 The Combined Condition

Often, these two conditions are combined into a single, comprehensive rule. A mapping f is a linear transformation if and only if:

$$f(ax + by) = af(x) + bf(y) \quad \text{for all } a, b \in F \text{ and } x, y \in U$$

This means that a linear transformation perfectly preserves linear combinations.

15.3 Verifying Linear Transformations

How do we prove if a given mapping is a linear transformation or not? We test the two conditions.

15.3.1 Example 1: A Valid Linear Transformation

Let

$$T : V_3(\mathbb{R}) \to V_2(\mathbb{R})$$

be defined by:

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_3)$$

Check Condition 1 (Addition): Let

$$x = (x_1, x_2, x_3)$$

and

$$y = (y_1, y_2, y_3).$$

Their sum is

$$x + y = (x_1 + y_1, x_2 + y_2, x_3 + y_3).$$

Apply T to the sum:

$$T(x + y) = ((x_1 + y_1) - (x_2 + y_2), (x_1 + y_1) - (x_3 + y_3))$$

Rearrange the terms inside the coordinates:

$$T(x + y) = (x_1 - x_2 + y_1 - y_2, x_1 - x_3 + y_1 - y_3)$$

Split the tuple into two separate tuples:

$$T(x + y) = (x_1 - x_2, x_1 - x_3) + (y_1 - y_2, y_1 - y_3)$$

Notice that the first part is exactly $T(x)$ and the second part is exactly $T(y)$:

$$T(x + y) = T(x) + T(y)$$

Condition 1 holds.

Check Condition 2 (Scalar Multiplication): Let $a \in \mathbb{R}$. The scaled vector is

$$ax = (ax_1, ax_2, ax_3).$$

Apply T:

$$T(ax) = (ax_1 - ax_2, ax_1 - ax_3)$$

Factor out the scalar a from both coordinates:

$$T(ax) = a(x_1 - x_2, x_1 - x_3)$$

This is exactly:

$$T(ax) = aT(x)$$

Condition 2 holds. Since both conditions are satisfied, T is a linear transformation.

15.3.2 Example 2: An Invalid Mapping

Consider

$$T : V_3(\mathbb{R}) \to \mathbb{R}$$

defined by:

$$T(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$$

Is this linear? No. Let's test the scalar multiplication condition.

$$T(ax) = T(ax_1, ax_2, ax_3) = (ax_1)^2 + (ax_2)^2 + (ax_3)^2 = a^2(x_1^2 + x_2^2 + x_3^2) = a^2T(x)$$

Because $a^2T(x) \neq aT(x)$, the mapping fails the scalar multiplication test. **Tip:** Mappings involving squares, cubes, constant additions (like $+5$), or multiplying coordinates together (like x_1x_2) are generally *not* linear transformations.

15.4 Algebraic Properties of Linear Transformations

Because vector spaces are abelian groups under addition, linear transformations inherit several elegant properties. Let $f : U \to V$ be a linear transformation.

15.4.1 Property 1: The Zero Vector Maps to the Zero Vector

$$f(\bar{0}) = \bar{0}'$$

(Where $\bar{0}$ is the zero vector of U and $\bar{0}'$ is the zero vector of V)

Proof: Let $\alpha \in U$. Then $f(\alpha) \in V$. Add the zero vector of V to $f(\alpha)$:

$$f(\alpha) + \bar{0}' = f(\alpha)$$

Now look at the domain U. Adding the zero vector changes nothing: $\alpha + \bar{0} = \alpha$. Apply f to both sides:

$$f(\alpha + \bar{0}) = f(\alpha)$$

Use the addition property of the linear transformation on the left side:

$$f(\alpha) + f(\bar{0}) = f(\alpha)$$

Equate our two expressions for $f(\alpha)$:

$$f(\alpha) + \bar{0}' = f(\alpha) + f(\bar{0})$$

By the cancellation law of groups, we cancel $f(\alpha)$ from both sides to get:

$$\bar{0}' = f(\bar{0})$$

15.4.2 Property 2: Inverses Map to Inverses

$$f(-\alpha) = -f(\alpha)$$

Proof: We know that in U:

$$\alpha + (-\alpha) = \bar{0}$$

Apply f to both sides:

$$f(\alpha + (-\alpha)) = f(\bar{0})$$

We just proved $f(\bar{0}) = \bar{0}'$. And we can split the left side using the addition property:

$$f(\alpha) + f(-\alpha) = \bar{0}'$$

This equation states that adding $f(-\alpha)$ to $f(\alpha)$ results in the zero vector. By the definition of an additive inverse in a group, $f(-\alpha)$ must be the inverse of $f(\alpha)$. Thus:

$$f(-\alpha) = -f(\alpha)$$

15.4.3 Property 3: Preservation of Subtraction

$$f(\alpha - \beta) = f(\alpha) - f(\beta)$$

Proof: Rewrite the subtraction as the addition of an inverse:

$$f(\alpha - \beta) = f(\alpha + (-\beta))$$

Use the addition property:

$$f(\alpha - \beta) = f(\alpha) + f(-\beta)$$

Substitute the inverse property ($f(-\beta) = -f(\beta)$):

$$f(\alpha - \beta) = f(\alpha) - f(\beta)$$

15.4.4 Property 4: Preservation of Generalized Linear Combinations

For any scalars $a_1, a_2, \ldots, a_n \in F$ and vectors $\alpha_1, \alpha_2, \ldots, \alpha_n \in U$:

$$f(a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n) = a_1 f(\alpha_1) + a_2 f(\alpha_2) + \cdots + a_n f(\alpha_n)$$

This states that the linear transformation property extends to combinations of any finite number of vectors. This can be formally proved using mathematical induction on n.

15.5 Key Takeaways

- A **Linear Transformation** is a function between vector spaces that preserves vector addition and scalar multiplication ($f(ax + by) = af(x) + bf(y)$).
- Formulas involving powers, constants, or coordinate multiplication are generally not linear.
- Linear transformations always map the zero vector of the domain to the zero vector of the co-domain ($f(\bar{0}) = \bar{0}'$).
- They preserve inverses and subtraction directly.

15.6 What Comes Next

We will continue analyzing linear transformations by exploring two critical subspaces associated with them: the **Kernel** (or Null Space) and the **Image Space**.

16 Lecture 16: Vector Space - Kernel and Range of a Homomorphism

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 16 of 27 **Video**: https://www.youtube.com/watch?v=NzuTxGGFcrg

16.1 Prerequisites

We know that a vector space homomorphism (or linear transformation) is a mapping $f : U \to V$ that preserves vector addition and scalar multiplication. It also maps the zero vector of the domain to the zero vector of the co-domain ($f(\bar{0}) = \bar{0}'$). Today, we define two critical sets associated with this mapping: the **Kernel** and the **Range**, and prove that they are both vector subspaces.

16.2 Definitions

16.2.1 1. The Kernel (Null Space)

The Kernel of a homomorphism $f : U \to V$, denoted as $\ker(f)$, K_f, or N_f, is the set of all vectors in the domain U that get mapped strictly to the zero vector of the co-domain V.

$$\ker(f) = \{\alpha \in U \mid f(\alpha) = \bar{0}'\}$$

Because $f(\bar{0}) = \bar{0}'$, the zero vector of U is *always* in the kernel. Thus, the kernel is never empty. The kernel lives entirely within the domain space U.

16.2.2 2. The Range Space

The Range of a homomorphism $f : U \to V$, denoted as R_f, is the set of all vectors in the co-domain V that are actual outputs (images) of the mapping.

$$R_f = \{\beta \in V \mid \beta = f(\alpha) \text{ for some } \alpha \in U\}$$

Because the zero vector maps to the zero vector, $\bar{0}'$ is always in the range. Thus, the range is never empty. The range lives entirely within the co-domain space V.

16.3 Theorem 1: The Kernel is a Subspace

Statement: Let $f : U \to V$ be a vector space homomorphism. The kernel of f, $\ker(f)$, is a subspace of the domain vector space U.

Proof: To prove a set is a subspace, we must prove that any linear combination of its elements also belongs to the set. First, we know $\ker(f) \neq \emptyset$ because $\bar{0} \in \ker(f)$.

Let α, β be two arbitrary vectors in $\ker(f)$. Because they are in the kernel, their images are zero:

$$f(\alpha) = \bar{0}' \quad \text{and} \quad f(\beta) = \bar{0}'$$

Let $a, b \in F$ be two scalars. We must check if the linear combination $a\alpha + b\beta$ is in the kernel. We test this by applying the mapping f:

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) \quad \text{(since } f \text{ is linear)}$$

Substitute our known values for $f(\alpha)$ and $f(\beta)$:

$$f(a\alpha + b\beta) = a(\bar{0}') + b(\bar{0}') = \bar{0}' + \bar{0}' = \bar{0}'$$

Because the image of the combination $(a\alpha + b\beta)$ is the zero vector, the combination itself must belong to the kernel.

$$a\alpha + b\beta \in \ker(f)$$

Therefore, $\ker(f)$ is a subspace of U.

16.4 Theorem 2: The Range is a Subspace

Statement: Let $f : U \to V$ be a vector space homomorphism. The range of f, R_f, is a subspace of the co-domain vector space V.

Proof: First, we know $R_f \neq \emptyset$ because $\bar{0}' = f(\bar{0})$, so $\bar{0}' \in R_f$.

Let β_1, β_2 be two arbitrary vectors in R_f. Because they are in the range, they must have pre-images in the domain U. Let's call their pre-images α_1 and α_2:

$$f(\alpha_1) = \beta_1 \quad \text{and} \quad f(\alpha_2) = \beta_2$$

Let $a, b \in F$ be two scalars. We must check if the linear combination $a\beta_1 + b\beta_2$ is in the range. To prove it is in the range, we must find a vector in U that maps to it. Let's guess the linear combination of the pre-images: $a\alpha_1 + b\alpha_2$. Because U is a vector space, $a\alpha_1 + b\alpha_2$ is definitely a valid vector in U. Let's apply f to it:

$$f(a\alpha_1 + b\alpha_2) = af(\alpha_1) + bf(\alpha_2) \quad \text{(since } f \text{ is linear)}$$

Substitute our known values for the images:

$$f(a\alpha_1 + b\alpha_2) = a\beta_1 + b\beta_2$$

This equation proves that $a\beta_1 + b\beta_2$ is the valid output of the input vector $(a\alpha_1 + b\alpha_2)$. Therefore, it belongs to the range.

$$a\beta_1 + b\beta_2 \in R_f$$

Therefore, R_f is a subspace of V.

16.5 Theorem 3: Injectivity and the Trivial Kernel

This is one of the most powerful theorems in linear algebra. It gives us a very easy way to check if a transformation is one-to-one.

Statement: Let $f : U \to V$ be a vector space homomorphism. The mapping f is one-to-one (injective) if and only if the kernel contains *only* the zero vector ($\ker(f) = \{\bar{0}\}$).

Proof: Since this is an "if and only if" theorem, we must prove both directions.

16.5.1 Part 1: Assume is one-to-one. Prove.

Let $\alpha \in \ker(f)$. By the definition of the kernel:

$$f(\alpha) = \bar{0}'$$

We also know a universal property of linear transformations:

$$f(\bar{0}) = \bar{0}'$$

We can equate the two expressions:

$$f(\alpha) = f(\bar{0})$$

Because we assumed f is one-to-one, distinct inputs must give distinct outputs. If the outputs are equal, the inputs *must* be equal:

$$\alpha = \bar{0}$$

Thus, any vector in the kernel must be the zero vector. $\ker(f) = \{\bar{0}\}$.

16.5.2 Part 2: Assume. Prove is one-to-one.

To prove f is one-to-one, we assume two outputs are equal and prove their inputs must be equal. Let $\alpha, \beta \in U$ such that:

$$f(\alpha) = f(\beta)$$

Subtract $f(\beta)$ from both sides:

$$f(\alpha) - f(\beta) = \bar{0}'$$

Use the subtraction property of linear transformations:

$$f(\alpha - \beta) = \bar{0}'$$

Because the image of the vector $(\alpha - \beta)$ is the zero vector, it must belong to the kernel:

$$\alpha - \beta \in \ker(f)$$

But our assumption is that the kernel *only* contains the zero vector. Therefore, this vector must be zero:

$$\alpha - \beta = \bar{0}$$

$$\alpha = \beta$$

Since $f(\alpha) = f(\beta)$ implies $\alpha = \beta$, the mapping is one-to-one. The theorem is proven in both directions.

16.6 Key Takeaways

- The **Kernel** ($\ker(f)$) is the set of inputs that get destroyed (mapped to zero). It is a subspace of the domain.
- The **Range** (R_f) is the set of all valid outputs. It is a subspace of the co-domain.
- A linear transformation is one-to-one if and only if its kernel is "trivial" (meaning $\ker(f) = \{\bar{0}\}$). If any non-zero vector sneaks into the kernel, the mapping is not one-to-one.

16.7 What Comes Next

We will use these concepts to define Isomorphisms (one-to-one and onto linear transformations) and study the Fundamental Theorem of Homomorphism.

17 Lecture 17: Vector Space - Isomorphism of Vector Spaces

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 17 of 27 **Video**: https://www.youtube.com/watch?v=KgZR5L9jTJA

17.1 Prerequisites

We know that a linear transformation (or homomorphism) preserves vector space structures (addition and scalar multiplication). Today, we look at the ultimate form of a linear transformation: an **Isomorphism**. When two spaces are isomorphic, they are algebraically identical.

17.2 What is an Isomorphism?

Let $U(F)$ and $V(F)$ be two vector spaces over the same field F. A mapping $f : U \to V$ is an **Isomorphism** if it satisfies three strict conditions:

1. **It is One-to-One (Injective)**: Every input gets a unique output. No two different vectors in U map to the same vector in V.
$$f(\alpha) = f(\beta) \implies \alpha = \beta$$
2. **It is Onto (Surjective)**: Every vector in the co-domain V is "hit" by the mapping. The range covers the entire co-domain. For every $\beta \in V$, there exists an $\alpha \in U$ such that $f(\alpha) = \beta$.
3. **It is a Linear Transformation (Homomorphism)**: It perfectly preserves linear combinations.
$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta)$$

If such a mapping exists between two spaces, we say the spaces are **Isomorphic** and we write:

$$U \cong V$$

When $U \cong V$, they are algebraically indistinguishable. They have the same dimension, the same number of basis elements, and their vectors behave exactly the same way under addition and scaling. The only difference is the *names* or *labels* given to the vectors.

17.3 Theorem 1: Isomorphism and Dimension

This theorem proves that for finite-dimensional spaces, "being isomorphic" is exactly the same as "having the same dimension."

Statement: Two finite-dimensional vector spaces over the same field are isomorphic if and only if they have the same dimension.
$$U \cong V \iff \dim(U) = \dim(V)$$

17.3.1 Part 1: Same Dimension Isomorphic

Let $\dim(U) = \dim(V) = n$. Let
$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$
be a basis for U. Let
$$S' = \{\beta_1, \beta_2, \ldots, \beta_n\}$$
be a basis for V.

We define a mapping $f : U \to V$ by matching their basis vectors:

$$f(a_1\alpha_1 + \cdots + a_n\alpha_n) = a_1\beta_1 + \cdots + a_n\beta_n$$

1. Prove f is One-to-One: Assume $f(\alpha) = f(\beta)$. Let

$$\alpha = a_1\alpha_1 + \cdots + a_n\alpha_n$$

and

$$\beta = b_1\alpha_1 + \cdots + b_n\alpha_n.$$

Substitute into the assumption:

$$a_1\beta_1 + \cdots + a_n\beta_n = b_1\beta_1 + \cdots + b_n\beta_n$$

Subtract to one side:

$$(a_1 - b_1)\beta_1 + \cdots + (a_n - b_n)\beta_n = \bar{0}'$$

Because S' is a basis, the β vectors are linearly independent. Thus, all coefficients are zero:

$$a_1 = b_1, \ldots, a_n = b_n \implies \alpha = \beta$$

Thus, f is one-to-one.
2. Prove f is Onto: Take any vector $\gamma \in V$. Since S' is a basis,

$$\gamma = c_1\beta_1 + \cdots + c_n\beta_n.$$

Because the coefficients c_i are in the field F, we can construct the vector

$$\alpha = c_1\alpha_1 + \cdots + c_n\alpha_n$$

in U. By our definition, $f(\alpha) = \gamma$. Thus, every vector in V has a pre-image. f is onto.
3. Prove f is Linear: Let $\alpha, \beta \in U$ and $k, l \in F$.

$$k\alpha + l\beta = (ka_1 + lb_1)\alpha_1 + \cdots + (ka_n + lb_n)\alpha_n$$

Apply f:

$$f(k\alpha + l\beta) = (ka_1 + lb_1)\beta_1 + \cdots + (ka_n + lb_n)\beta_n$$

Split the coefficients:

$$f(k\alpha + l\beta) = k(a_1\beta_1 + \cdots + a_n\beta_n) + l(b_1\beta_1 + \cdots + b_n\beta_n) = kf(\alpha) + lf(\beta)$$

Thus, f is linear. Since f is one-to-one, onto, and linear, $U \cong V$.

17.3.2 Part 2: Isomorphic Same Dimension

Assume $U \cong V$. This means there exists an isomorphism $f : U \to V$. Let $\dim(U) = n$, and let

$$S = \{\alpha_1, \ldots, \alpha_n\}$$

be its basis. We map the basis S to V to create a new set:

$$S' = \{f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_n)\}$$

1. Prove S' is Linearly Independent: Set a linear combination to zero:

$$a_1 f(\alpha_1) + \cdots + a_n f(\alpha_n) = \bar{0}'$$

Because f is linear, we can pull f outside:

$$f(a_1\alpha_1 + \cdots + a_n\alpha_n) = \bar{0}'$$

We know $f(\bar{0}) = \bar{0}'$. Since f is one-to-one, equal images mean equal inputs:

$$a_1\alpha_1 + \cdots + a_n\alpha_n = \bar{0}$$

Because S is a basis of U, it is independent, so all coefficients a_i must be zero. Thus, S' is independent.
2. Prove S' Spans V: Take any $\gamma \in V$. Because f is onto, there exists $\alpha \in U$ such that $f(\alpha) = \gamma$. Since S is a basis of U,

$$\alpha = c_1\alpha_1 + \cdots + c_n\alpha_n.$$

Apply f:

$$\gamma = f(\alpha) = f(c_1\alpha_1 + \cdots + c_n\alpha_n) = c_1 f(\alpha_1) + \cdots + c_n f(\alpha_n)$$

This proves any vector $\gamma \in V$ can be built using S'. Thus, S' is a basis for V. Since S' contains n elements, $\dim(V) = n$. Therefore, $\dim(U) = \dim(V)$.

17.4 Theorem 2: The Universal Isomorphism

This theorem states that *every* vector space of dimension n is just the standard tuple space $V_n(F)$ in disguise.

Statement: Every n-dimensional vector space $V(F)$ is isomorphic to the space $V_n(F)$ (the space of n-tuples) over the same field F.

$$V(F) \cong V_n(F)$$

Proof Sketch: Let

$$S = \{\alpha_1, \ldots, \alpha_n\}$$

be a basis for V. Any vector $\alpha \in V$ can be uniquely written as

$$\alpha = a_1\alpha_1 + \cdots + a_n\alpha_n.$$

Define a mapping f that takes this vector and maps it to a simple n-tuple of its coordinates:

$$f(a_1\alpha_1 + \cdots + a_n\alpha_n) = (a_1, a_2, \ldots, a_n)$$

Because coordinates relative to a basis are unique, this mapping is well-defined, one-to-one, onto, and perfectly linear. Thus, any n-dimensional space behaves exactly like the space of n-tuples.

17.5 Key Takeaways

- An **Isomorphism** is a linear transformation that is both one-to-one and onto.
- If two finite-dimensional spaces share the same field and the same dimension, they are mathematically identical ($U \cong V$).
- An isomorphism maps a basis of the domain to a basis of the co-domain.

17.6 What Comes Next

We will conclude Phase 6 with the Fundamental Theorem of Homomorphism for vector spaces.

18 Lecture 18: Vector Space - Fundamental Theorem of Vector Space Homomorphism

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 18 of 27 **Video**: https://www.youtube.com/watch?v=jAH5rcWIyCE

18.1 Prerequisites

We know what a quotient space V/W is, and we know that a homomorphism (linear transformation) T has a kernel, which is a subspace of the domain. Today, we connect these two concepts in one of the most important theorems in linear algebra.

Before tackling the main theorem, we prove a preliminary lemma that explains how a quotient space interacts with its parent vector space.

18.2 Lemma: The Natural Projection

Statement: If $V(F)$ is a vector space and W is its subspace, then the quotient space V/W is a homomorphic image of $V(F)$ with kernel W.

Proof: We must define a mapping from V to V/W, prove it is a homomorphism, and show its kernel is exactly W.

1. Define the Mapping: Let $f : V \to V/W$ be defined by mapping every vector in V to its corresponding coset in V/W:

$$f(\alpha) = W + \alpha \quad \text{for all } \alpha \in V$$

2. Verify it is Well-Defined: Let $\alpha, \beta \in V$. If $\alpha = \beta$, then adding the subspace W to both yields $W + \alpha = W + \beta$. Therefore, $f(\alpha) = f(\beta)$. The mapping is perfectly consistent.

3. Verify it is a Homomorphism (Linear): Let $\alpha, \beta \in V$ and $a, b \in F$.

$$f(a\alpha + b\beta) = W + (a\alpha + b\beta)$$

Using the rules of coset addition and scalar multiplication in the quotient space:

$$f(a\alpha + b\beta) = (W + a\alpha) + (W + b\beta) = a(W + \alpha) + b(W + \beta)$$

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta)$$

Thus, f is a homomorphism.

4. Find its Kernel: The kernel of f consists of all vectors in V that map to the identity element of V/W. The identity element of V/W is the subspace W.

$$\ker(f) = \{\alpha \in V \mid f(\alpha) = W\}$$

By definition of f:

$$W + \alpha = W$$

We know from quotient space properties that $W + \alpha = W \iff \alpha \in W$. Thus, any vector in the kernel must belong to W, and any vector in W belongs to the kernel.

$$\ker(f) = W$$

The lemma is proven.

18.3 The Fundamental Theorem of Homomorphism

This theorem states that if you take any linear mapping, you can "mod out" (divide out) the information destroyed by the mapping (the kernel) to create a perfect one-to-one correspondence (an isomorphism) with the image.

Statement: Let $V(F)$ and $U(F)$ be vector spaces, and let $T : V \to U$ be an onto linear transformation (homomorphism) with kernel W. Then:

$$V/W \cong U$$

(If T is not onto, the theorem is simply $V/W \cong Im(T)$).

Proof: Because W is the kernel of T, it is a valid subspace of V. Thus, the quotient space V/W exists. We must construct a new mapping $f : V/W \to U$ and prove it is an isomorphism (well-defined, one-to-one, onto, linear).

1. Define the Mapping: For any coset $W + \alpha \in V/W$, we define its image under f to be the image of its representative vector under T:

$$f(W + \alpha) = T(\alpha)$$

2. Verify it is Well-Defined: Because a single coset can be represented by many different vectors (e.g., $W + \alpha = W + \beta$), we must ensure our mapping gives the same output regardless of the representative chosen. Assume two cosets are equal:

$$W + \alpha = W + \beta$$

This implies $\alpha - \beta \in W$. Because W is the kernel of T, any vector in W maps to zero under T:

$$T(\alpha - \beta) = \bar{0}'$$

Since T is linear:

$$T(\alpha) - T(\beta) = \bar{0}' \implies T(\alpha) = T(\beta)$$

By the definition of f:

$$f(W + \alpha) = f(W + \beta)$$

Equal inputs yield equal outputs. The mapping is well-defined.

3. Verify it is One-to-One: We just reverse the well-definedness steps. Assume the outputs are equal:

$$f(W + \alpha) = f(W + \beta)$$

$$T(\alpha) = T(\beta) \implies T(\alpha) - T(\beta) = \bar{0}' \implies T(\alpha - \beta) = \bar{0}'$$

Because the image of $(\alpha - \beta)$ under T is zero, it belongs to the kernel W:

$$\alpha - \beta \in W$$

This implies that their cosets are equal:

$$W + (\alpha - \beta) = W \implies W + \alpha = W + \beta$$

Equal outputs imply equal inputs. The mapping is one-to-one.

4. Verify it is Onto: Because we assumed the original mapping $T : V \to U$ is onto, for every vector $u \in U$, there exists some $\alpha \in V$ such that $T(\alpha) = u$. For this vector α, the coset $W + \alpha \in V/W$ satisfies:

$$f(W + \alpha) = T(\alpha) = u$$

Thus, every element in U is hit by f. The mapping is onto.

5. Verify it is Linear: Let $W + \alpha, W + \beta \in V/W$ and $a, b \in F$.

$$f(a(W + \alpha) + b(W + \beta)) = f((W + a\alpha) + (W + b\beta)) = f(W + (a\alpha + b\beta))$$

By the definition of f:

$$= T(a\alpha + b\beta)$$

Because T is linear:

$$= aT(\alpha) + bT(\beta)$$

Substitute the definition of f back in:

$$= af(W + \alpha) + bf(W + \beta)$$

Thus, f is a linear transformation.

Since f is well-defined, one-to-one, onto, and linear, it is an isomorphism.

$$V/W \cong U$$

18.4 Key Takeaways

- The **Fundamental Theorem of Homomorphism** bridges vector spaces, quotient spaces, and linear transformations.
- By forming a quotient space V/W where $W = \ker(T)$, we effectively "collapse" everything that gets mapped to zero into a single identity element. This removes all redundancy, turning the mapping into a perfect one-to-one correspondence (an isomorphism) with the image.

18.5 What Comes Next

With the theoretical framework of linear transformations complete, we will move to Phase 7, where we ground these concepts in computation by investigating **Matrix Representations** and the **Rank-Nullity Theorem**.

19 Lecture 19: Vector Space - Matrix Representation of Linear Transformation

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 19 of 27 **Video**: https://www.youtube.com/watch?v=NCsoLIlz_lU

19.1 Prerequisites

We know that linear transformations map vectors from one space to another. But working with abstract mappings is difficult. Today, we bridge the gap between abstract vector spaces and computational linear algebra by representing these transformations as **Matrices**. Once a transformation is expressed as a matrix, operations like evaluating images, finding compositions, or finding inverses become simple matrix multiplications.

19.2 Theory of Matrix Representation

Let $U(F)$ and $V(F)$ be two vector spaces over the field F. Let $T : U \to V$ be a linear transformation. To represent T as a matrix, we must fix an **ordered basis** for both spaces: * Let

$$B = \{u_1, u_2, \dots, u_n\}$$

be the ordered basis for U. * Let

$$B' = \{v_1, v_2, \dots, v_m\}$$

be the ordered basis for V.

19.2.1 The Process

1. **Find the Images of the Domain Basis**: We take each basis vector of U (that is, each $u_j \in B$) and find its image under T, which is $T(u_j)$. This image lives in V.
2. **Express Images using the Co-Domain Basis**: Since $T(u_j) \in V$, we can write it uniquely as a linear combination of the basis vectors in B':

$$\begin{gathered}
T(u_1) = a_{11}v_1 + a_{12}v_2 + \cdots + a_{1m}v_m \\
T(u_2) = a_{21}v_1 + a_{22}v_2 + \cdots + a_{2m}v_m \\
\vdots \\
T(u_n) = a_{n1}v_1 + a_{n2}v_2 + \cdots + a_{nm}v_m
\end{gathered}$$

3. **Construct the Matrix**: We take the scalar coefficients from each linear combination and place them as **columns** in a matrix. The resulting matrix is the matrix representation of T relative to the bases B and B', denoted as $[T]_B^{B'}$.

$$[T]_B^{B'} = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \dots & a_{nm} \end{bmatrix}$$

(Note: If $T : V \to V$ is an operator mapping a space to itself, we usually use the same basis for both sides, denoted simply as $[T]_B$.)

19.3 Standard Bases

If the problem does not explicitly provide a basis, we assume the **standard basis**. * For $V_2(\mathbb{R})$, the standard basis is $B = \{(1, 0), (0, 1)\}$. * For $V_3(\mathbb{R})$, the standard basis is $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

19.4 Numerical Example 1

Problem: Find the matrix of the linear transformation

$$T : V_3(\mathbb{R}) \to V_3(\mathbb{R})$$

defined by:

$$T(a, b, c) = (2b + c, a - 4b, 3a)$$

with respect to the standard basis B.

Solution: The standard basis is

$$B = \{e_1(1, 0, 0), e_2(0, 1, 0), e_3(0, 0, 1)\}.$$

Step 1: Compute Images of Basis Vectors * $T(1, 0, 0) = (2(0) + 0, 1 - 4(0), 3(1)) = (0, 1, 3)$ * $T(0, 1, 0) = (2(1) + 0, 0 - 4(1), 3(0)) = (2, -4, 0)$ * $T(0, 0, 1) = (2(0) + 1, 0 - 4(0), 3(0)) = (1, 0, 0)$

Step 2: Express as Linear Combinations Because we are using the standard basis, the coordinates are simply the components of the vectors: * $(0, 1, 3) = 0(1, 0, 0) + 1(0, 1, 0) + 3(0, 0, 1)$ * $(2, -4, 0) = 2(1, 0, 0) - 4(0, 1, 0) + 0(0, 0, 1)$ * $(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1)$

Step 3: Compile the Matrix Write these coordinates as columns:

$$[T]_B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -4 & 0 \\ 3 & 0 & 0 \end{bmatrix}$$

19.5 Numerical Example 2 (Non-Standard Basis)

Problem: Find the matrix of the *same* linear transformation T from Example 1, but this time with respect to the non-standard basis:

$$B' = \{v_1(1, 1, 1), v_2(1, 1, 0), v_3(1, 0, 0)\}$$

Solution: Step 1: Derive the General Coordinate Formula To express any vector (a, b, c) as a combination of B':

$$(a, b, c) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0) = (x + y + z, x + y, x)$$

Comparing components: 1. $x = c$ 2. $x + y = b \implies y = b - c$ 3. $x + y + z = a \implies z = a - b$ Coordinate formula: $x = c, \quad y = b - c, \quad z = a - b$

Step 2: Compute Images and their Coordinates * **For

$$v_1 = (1, 1, 1)$$

:** $T(1, 1, 1) = (3, -3, 3)$. Using the formula $(a = 3, b = -3, c = 3)$: $x = 3$, $y = -3 - 3 = -6$, $z = 3 - (-3) = 6$.

$$T(v_1) = 3v_1 - 6v_2 + 6v_3$$

- **For

$$v_2 = (1, 1, 0)$$

:** $T(1, 1, 0) = (2, -3, 3)$. Using the formula $(a = 2, b = -3, c = 3)$: $x = 3$, $y = -3 - 3 = -6$, $z = 2 - (-3) = 5$.

$$T(v_2) = 3v_1 - 6v_2 + 5v_3$$

- **For

$$v_3 = (1, 0, 0)$$

:** $T(1, 0, 0) = (0, 1, 3)$. Using the formula $(a = 0, b = 1, c = 3)$: $x = 3$, $y = 1 - 3 = -2$, $z = 0 - 1 = -1$.

$$T(v_3) = 3v_1 - 2v_2 - 1v_3$$

Step 3: Compile the Matrix Write the computed coordinates as columns:

$$[T]_{B'} = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix}$$

19.5.1 Part 3: Verification of the Relation

Let's verify the relation $[T]_{B'}[\alpha]_{B'} = [T(\alpha)]_{B'}$ for any vector $\alpha = (a, b, c) \in V_3(\mathbb{R})$.

1. Express $[\alpha]_{B'}$ in Matrix Form Using our derived coordinate formula ($x = c, y = b - c, z = a - b$):

$$[\alpha]_{B'} = \begin{bmatrix} c \\ b-c \\ a-b \end{bmatrix}$$

2. Express $[T(\alpha)]_{B'}$ in Matrix Form We know $T(\alpha) = (2b+c, a-4b, 3a)$. Let $a' = 2b+c$, $b' = a-4b$, and $c' = 3a$. Applying the coordinate formula ($x = c', y = b'-c', z = a'-b'$): * $x = 3a$ * $y = (a-4b)-3a = -2a-4b$ * $z = (2b+c) - (a-4b) = -a + 6b + c$

$$[T(\alpha)]_{B'} = \begin{bmatrix} 3a \\ -2a-4b \\ -a+6b+c \end{bmatrix}$$

3. Compute the Matrix Product

$$[T]_{B'}[\alpha]_{B'} = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix} \begin{bmatrix} c \\ b-c \\ a-b \end{bmatrix}$$

$$= \begin{bmatrix} 3c + 3(b-c) + 3(a-b) \\ -6c - 6(b-c) - 2(a-b) \\ 6c + 5(b-c) - (a-b) \end{bmatrix}$$

$$= \begin{bmatrix} 3c + 3b - 3c + 3a - 3b \\ -6c - 6b + 6c - 2a + 2b \\ 6c + 5b - 5c - a + b \end{bmatrix}$$

$$= \begin{bmatrix} 3a \\ -2a-4b \\ -a+6b+c \end{bmatrix}$$

Since $[T]_{B'}[\alpha]_{B'} = [T(\alpha)]_{B'}$, the relation is verified and proved. This demonstrates that the matrix perfectly encapsulates the transformation!

19.6 Numerical Example 3 (Different Dimensions)

Problem: Find the matrix of

$$T : \mathbb{R}^3 \to \mathbb{R}^2$$

defined by:

$$T(x, y, z) = (2x - 4y + 9z, 5x + 3y - 2z)$$

with respect to the standard bases.

Solution: Domain basis $B = \{(1,0,0), (0,1,0), (0,0,1)\}$. Co-domain basis $B' = \{(1,0), (0,1)\}$. Compute images: * $T(1,0,0) = (2(1) - 0 + 0, 5(1) + 0 - 0) = (2, 5)$ * $T(0,1,0) = (0 - 4(1) + 0, 0 + 3(1) - 0) = (-4, 3)$ * $T(0,0,1) = (0 - 0 + 9(1), 0 + 0 - 2(1)) = (9, -2)$

Since B' is standard, these components *are* the coordinates. Write them as columns:

$$[T]_B^{B'} = \begin{bmatrix} 2 & -4 & 9 \\ 5 & 3 & -2 \end{bmatrix}$$

(Notice that a transformation from $\mathbb{R}^3 \to \mathbb{R}^2$ yields a 2×3 matrix.)

19.7 Key Takeaways

- A linear transformation is completely determined by its action on a basis.
- To build a transformation matrix, find the images of the domain basis, express them as coordinates relative to the co-domain basis, and stack those coordinates as **columns**.
- The matrix perfectly encapsulates the transformation:

$$A\vec{x} = \vec{y}.$$

19.8 What Comes Next

We will explore the relationship between the matrix representation and the **Rank and Nullity** of a linear transformation, leading into the Rank-Nullity Theorem.

20 Lecture 20: Vector Space - Inverse of Linear Transformation

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 20 of 27 **Video**: https://www.youtube.com/watch?v=bXeQILCHnVY

20.1 Prerequisites

We know how to represent a linear transformation $T : V \to W$ as a matrix $[T]$. Today, we will learn how to reverse this mapping. Finding the **Inverse Linear Transformation** T^{-1} means finding a mapping that takes vectors from the co-domain back to their original vectors in the domain.

20.2 Condition for Invertibility

Let $T : V \to W$ be a linear transformation. For T to have an inverse $T^{-1} : W \to V$, the transformation must be a perfect one-to-one correspondence (an isomorphism). Practically, when working with matrix representations, this means:

1. **Square Matrix Requirement**: The dimension of the domain must equal the dimension of the co-domain. For a transformation $T : V \to V$, its matrix

$$A = [T]_B$$

will be a square matrix. 2. **Non-Singularity**: The determinant of the transformation matrix must be non-zero.

$$|A| \neq 0$$

If $|A| \neq 0$, the linear transformation is called **non-singular** and is guaranteed to be invertible. If

$$|A| = 0$$

, it is singular and cannot be inverted (information was lost, meaning the kernel is not just the zero vector).

20.3 Finding the Inverse Transformation

If the transformation is invertible, its inverse is perfectly represented by the inverse of its matrix.

20.3.1 The Steps

1. **Construct the Matrix**: Find the matrix

$$A = [T]_B$$

of the transformation with respect to a given basis (usually the standard basis). 2. **Verify Non-Singularity**: Compute the determinant $|A|$. Ensure $|A| \neq 0$. 3. **Compute the Inverse Matrix**: Use the standard formula for matrix inversion:

$$A^{-1} = \frac{\operatorname{adj}(A)}{|A|}$$

`This inverse matrix *is* the matrix representation of the inverse transformation:`

$$[T^{-1}]_B = A^{-1}$$

4. **Derive the Formula**: Multiply the inverse matrix by a general column vector to find the explicit formula for T^{-1}:

$$T^{-1}(x, y, z) = A^{-1} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

20.4 Numerical Example 1

Problem: Find the inverse linear transformation T^{-1} for the transformation

$$T : \mathbb{R}^3 \to \mathbb{R}^3$$

defined by:

$$T(x, y, z) = (2x + y + z, x + y + 2z, x - 2z)$$

with respect to the standard basis.

Step 1: Construct the Matrix A Use the standard basis $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Compute the images: * $T(1, 0, 0) = (2(1)+0+0, 1+0+0, 1-0) = (2, 1, 1)$ * $T(0, 1, 0) = (0+1+0, 0+1+0, 0-0) = (1, 1, 0)$ * $T(0, 0, 1) = (0 + 0 + 1, 0 + 0 + 2(1), 0 - 2(1)) = (1, 2, -2)$

Write these coordinate vectors as columns to form the matrix A:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix}$$

Step 2: Verify Non-Singularity Compute the determinant of A:

$$|A| = 2 \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} + 1 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$$

$$|A| = 2(-2) - 1(-4) + 1(-1) = -4 + 4 - 1 = -1$$

Since

$$|A| = -1 \neq 0$$

, the transformation is non-singular and invertible.

Step 3: Compute the Inverse Matrix A^{-1} Find the adjoint matrix by computing all 9 cofactors A_{ij}:

- $A_{11} = (-1)^{1+1}(1(-2) - 0) = -2$
- $A_{12} = (-1)^{1+2}(1(-2) - 2) = 4$
- $A_{13} = (-1)^{1+3}(0 - 1) = -1$
- $A_{21} = (-1)^{2+1}(1(-2) - 0) = 2$
- $A_{22} = (-1)^{2+2}(2(-2) - 1) = -5$
- $A_{23} = (-1)^{2+3}(0 - 1) = 1$
- $A_{31} = (-1)^{3+1}(2 - 1) = 1$
- $A_{32} = (-1)^{3+2}(4 - 1) = -3$
- $A_{33} = (-1)^{3+3}(2 - 1) = 1$

$$\text{adj}(A) = \begin{bmatrix} -2 & 2 & 1 \\ 4 & -5 & -3 \\ -1 & 1 & 1 \end{bmatrix}$$

Calculate the inverse:

$$A^{-1} = \frac{1}{|A|}\text{adj}(A) = \frac{1}{-1} \begin{bmatrix} -2 & 2 & 1 \\ 4 & -5 & -3 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -1 \\ -4 & 5 & 3 \\ 1 & -1 & -1 \end{bmatrix}$$

Step 4: Derive the Formula Multiply the inverse matrix by $(x, y, z)^T$:

$$T^{-1}(x, y, z) = \begin{bmatrix} 2 & -2 & -1 \\ -4 & 5 & 3 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Extracting the rows gives the final formula:

$$T^{-1}(x, y, z) = (2x - 2y - z, -4x + 5y + 3z, x - y - z)$$

Verification: We know $T(1,0,0) = (2,1,1)$. Let's plug $(2,1,1)$ into our new formula.

$$T^{-1}(2,1,1) = (2(2) - 2(1) - 1, -4(2) + 5(1) + 3(1), 2 - 1 - 1)$$

$$T^{-1}(2,1,1) = (4 - 2 - 1, -8 + 5 + 3, 0) = (1,0,0)$$

The formula perfectly reverses the transformation!

20.5 Numerical Example 2 (Homework)

Problem: Let

$$T : \mathbb{R}^3 \to \mathbb{R}^3$$

be a linear transformation defined by:

$$T(x,y,z) = (3x + z, -2x + y, -x + 2y + 4z)$$

Prove that T is invertible, and find the formula for T^{-1}.

Solution Outline: 1. Matrix representation: Compute $T(1,0,0) = (3,-2,-1)$, $T(0,1,0) = (0,1,2)$, $T(0,0,1) = (1,0,4)$.

$$A = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}$$

2. Determinant:

$$|A| = 3(4 - 0) - 0 + 1(-4 + 1) = 12 - 3 = 9 \neq 0$$

Because the determinant is 9 (non-zero), T is invertible.

3. Adjoint and Inverse: Calculating the cofactors gives the adjoint:

$$\text{adj}(A) = \begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{9} \begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}$$

4. Final Formula:

$$T^{-1}(x,y,z) = \frac{1}{9}(4x + 2y - z, 8x + 13y - 2z, -3x - 6y + 3z)$$

20.6 Key Takeaways

- A linear transformation is invertible (non-singular) if and only if the determinant of its matrix representation is non-zero ($|A| \neq 0$).
- The matrix of the inverse transformation is exactly the inverse of the original transformation's matrix (

$$[T^{-1}]_B = [T]_B^{-1}$$

). * Matrix representation turns the complex algebraic problem of reversing a function into the standardized mechanical process of finding a matrix inverse.

21 Lecture 21: Vector Space - Rank-Nullity Theorem

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 21 of 27 **Video**: https://www.youtube.com/watch?v=HnFFuXDi9p8

21.1 Prerequisites

We know that a linear transformation $T : U \to V$ comes with two critical subspaces: the **Range Space** ($R(T)$, living in V) and the **Null Space** ($N(T)$ or Kernel, living in U). Today, we will study one of the most celebrated theorems in linear algebra, which reveals a deep dimensional relationship between these two spaces and the domain itself.

21.2 Definitions: Rank and Nullity

Let $T : U \to V$ be a linear transformation, and let U be a finite-dimensional vector space.

21.2.1 1. The Rank

The range space $R(T)$ is a subspace of V. Because it is a subspace, it has a dimension. The **Rank** of T, denoted as $\rho(T)$, is simply the dimension of its range space.

$$\text{Rank}(T) = \rho(T) = \dim(R(T))$$

21.2.2 2. The Nullity

The null space $N(T)$ is a subspace of U. Because it is a subspace, it also has a dimension. The **Nullity** of T, denoted as $\nu(T)$, is the dimension of its null space.

$$\text{Nullity}(T) = \nu(T) = \dim(N(T))$$

Conceptually: The nullity measures how much "information" the transformation crushes to zero, while the rank measures the number of actual "dimensions" the transformation preserves in the output.

21.3 The Rank-Nullity Theorem

Statement: Let $T : U \to V$ be a linear transformation from a finite-dimensional vector space U into a vector space V. Then, the sum of the rank and the nullity of T is equal to the dimension of the domain U.

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(U)$$

21.3.1 Proof

Let $\dim(U) = n$. Because $N(T)$ is a subspace of U, its dimension must be less than or equal to n. Let its dimension be k:

$$\dim(N(T)) = k \quad (k \le n)$$

Let

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_k\}$$

be a basis for the null space $N(T)$. Because S is a basis, it is a linearly independent set within U. By the **Basis Extension Theorem**, any linearly independent set can be extended to form a basis of the entire vector space. Let's extend S to form a basis S' for the entire domain U:

$$S' = \{\alpha_1, \alpha_2, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_n\}$$

Our goal is to prove that the rank (the dimension of $R(T)$) is exactly $n - k$. To do this, we will prove that the images of the *extended* basis vectors:

$$B' = \{T(\alpha_{k+1}), T(\alpha_{k+2}), \dots, T(\alpha_n)\}$$

form a basis for $R(T)$. We must show they span $R(T)$ and are linearly independent.

21.3.1.1 Step 1: Prove Spans

Let $\beta \in R(T)$. By definition, there exists an $\alpha \in U$ such that $T(\alpha) = \beta$. Because S' is a basis for U, we can express α as a linear combination of S':

$$\alpha = a_1\alpha_1 + \cdots + a_k\alpha_k + a_{k+1}\alpha_{k+1} + \cdots + a_n\alpha_n$$

Apply T to both sides:

$$\beta = T(a_1\alpha_1 + \cdots + a_k\alpha_k + a_{k+1}\alpha_{k+1} + \cdots + a_n\alpha_n)$$

Because T is linear, we can pull the scalars out:

$$\beta = a_1T(\alpha_1) + \cdots + a_kT(\alpha_k) + a_{k+1}T(\alpha_{k+1}) + \cdots + a_nT(\alpha_n)$$

But wait! The vectors $\alpha_1, \ldots, \alpha_k$ belong to the null space. This means their images $T(\alpha_i)$ are all the zero vector ($\bar{0}'$). The first k terms completely vanish:

$$\beta = a_1(\bar{0}') + \cdots + a_k(\bar{0}') + a_{k+1}T(\alpha_{k+1}) + \cdots + a_nT(\alpha_n)$$

$$\beta = a_{k+1}T(\alpha_{k+1}) + \cdots + a_nT(\alpha_n)$$

This proves that *any* vector $\beta \in R(T)$ can be built using only the vectors in B'. Therefore, B' spans $R(T)$.

21.3.1.2 Step 2: Prove is Linearly Independent

Set a linear combination of B' equal to the zero vector of V:

$$b_{k+1}T(\alpha_{k+1}) + \cdots + b_nT(\alpha_n) = \bar{0}'$$

Because T is linear, we pull T outside:

$$T(b_{k+1}\alpha_{k+1} + \cdots + b_n\alpha_n) = \bar{0}'$$

This equation states that the image of the vector

$$(b_{k+1}\alpha_{k+1} + \cdots + b_n\alpha_n)$$

is the zero vector. By definition, this vector *must* belong to the null space $N(T)$:

$$b_{k+1}\alpha_{k+1} + \cdots + b_n\alpha_n \in N(T)$$

Because it is in the null space, it can be written as a linear combination of the null space's basis S:

$$b_{k+1}\alpha_{k+1} + \cdots + b_n\alpha_n = c_1\alpha_1 + \cdots + c_k\alpha_k$$

Bring all terms to one side:

$$-c_1\alpha_1 - \cdots - c_k\alpha_k + b_{k+1}\alpha_{k+1} + \cdots + b_n\alpha_n = \bar{0}$$

Look at the vectors in this equation: they are exactly the basis vectors of S'. Because S' is a basis for U, it is a linearly independent set. The *only* way a linear combination of linearly independent vectors equals zero is if every single scalar is zero.

$$-c_1 = \cdots = -c_k = 0 \quad \text{and} \quad b_{k+1} = \cdots = b_n = 0$$

Because

$$b_{k+1} = \cdots = b_n = 0$$

, the set B' is linearly independent.

21.3.1.3 Step 3: Conclusion

Since B' spans $R(T)$ and is linearly independent, it is a basis for $R(T)$. Count the vectors in B': there are $(n-k)$ vectors.

$$\dim(R(T)) = n - k$$

By definition:

$$\text{Rank}(T) = n - k$$

Finally, add the rank and the nullity:

$$\text{Rank}(T) + \text{Nullity}(T) = (n-k) + k = n$$

Since $n = \dim(U)$, the theorem is proven:

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(U)$$

21.4 Key Takeaways

- The **Rank-Nullity Theorem** proves that the dimensions of the domain are strictly conserved across a linear mapping.
- If a transformation crushes a lot of dimensions to zero (high nullity), it must result in a "flattened" output with fewer dimensions (low rank). The total dimensions are always preserved.

21.5 What Comes Next

We will explore the implications of this theorem, define singular and non-singular transformations more formally, and solve numerical problems using the rank-nullity relation.

22 Lecture 22: Vector Space - Invertible, Singular & Nonsingular Linear Transformations

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 22 of 27 **Video**: https://www.youtube.com/watch?v=qIOCtAR0Qw0

22.1 Prerequisites

We know how to calculate the inverse of a linear transformation using matrix representations. Today, we formalize the algebraic definitions of invertibility, singularity, and the composition of linear transformations, proving several foundational theorems.

22.2 1. Invertible Linear Transformations

22.2.1 Definition

A linear transformation $T : U \to V$ is said to be **invertible** if it is a bijective mapping. That means T must satisfy three conditions: 1. **Linearity**: It preserves addition and scalar multiplication. 2. **One-to-One (Injective)**: $T(\alpha) = T(\beta) \implies \alpha = \beta$. 3. **Onto (Surjective)**: For every $\beta \in V$, there exists an $\alpha \in U$ such that $T(\alpha) = \beta$.

If these conditions are met, an inverse mapping $T^{-1} : V \to U$ exists, defined by:

$$T^{-1}(\beta) = \alpha \iff T(\alpha) = \beta$$

22.2.2 Theorem 1: Linearity of the Inverse

Statement: If $T : V_1 \to V_2$ is an invertible linear transformation, then its inverse $T^{-1} : V_2 \to V_1$ is also a linear transformation.

Proof: We must prove that T^{-1} preserves linear combinations:

$$T^{-1}(a\alpha' + b\beta') = aT^{-1}(\alpha') + bT^{-1}(\beta').$$

Let $\alpha', \beta' \in V_2$. Because T is onto, they have unique pre-images $\alpha, \beta \in V_1$:

$$T(\alpha) = \alpha' \implies T^{-1}(\alpha') = \alpha$$

$$T(\beta) = \beta' \implies T^{-1}(\beta') = \beta$$

Let $a, b \in F$. Consider the linear combination $a\alpha + b\beta$ in V_1. Apply T:

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) \quad \text{(since } T \text{ is linear)}$$

Substitute the images:

$$T(a\alpha + b\beta) = a\alpha' + b\beta'$$

Apply the inverse mapping T^{-1} to both sides:

$$a\alpha + b\beta = T^{-1}(a\alpha' + b\beta')$$

Substitute back the inverse expressions for α and β:

$$aT^{-1}(\alpha') + bT^{-1}(\beta') = T^{-1}(a\alpha' + b\beta')$$

Thus, T^{-1} is a linear transformation.

22.2.3 Result 2: Composition with the Inverse

If $T : U \to V$ is an invertible linear transformation, then composing it with its inverse yields the identity transformation.

$$TT^{-1} = I_V \quad \text{and} \quad T^{-1}T = I_U$$

(Where I_V and I_U are the identity mappings that do nothing to the vectors in V and U respectively.)

Proof: 1. Let $\beta \in V$. Since T is invertible, let $T^{-1}(\beta) = \alpha \in U \implies T(\alpha) = \beta$.

```
$$
(T T^{-1})(\beta) = T(T^{-1}(\beta)) = T(\alpha) = \beta = I_V(\beta)
Since this holds for all $\beta \in V$, we have $T T^{-1} = I_V$.
```

2. Let $\alpha \in U$. Let $T(\alpha) = \beta \in V \implies T^{-1}(\beta) = \alpha$.

$$(T^{-1}T)(\alpha) = T^{-1}(T(\alpha)) = T^{-1}(\beta) = \alpha = I_U(\alpha)$$

Since this holds for all $\alpha \in U$, we have $T^{-1}T = I_U$.

22.3 2. Singular and Nonsingular Transformations

Let $T : U \to V$ be a linear transformation. Let $\bar{0}$ and $\bar{0}'$ be the zero vectors of U and V, respectively.

22.3.1 Definitions

1. Singular Transformation: T is singular if it maps at least one non-zero vector to the zero vector. There exists some $\alpha \neq \bar{0}$ such that $T(\alpha) = \bar{0}'$. *(This means information is lost, and the mapping cannot be reversed.)*

2. Nonsingular Transformation: T is nonsingular if the *only* vector that maps to the zero vector is the zero vector itself.

$$T(\alpha) = \bar{0}' \implies \alpha = \bar{0}$$

(This means the null space (kernel) is trivial: $N(T) = \{\bar{0}\}$.)

22.3.2 Theorem 2: Nonsingularity and One-to-One

Statement: A linear transformation $T : U \to V$ is nonsingular if and only if T is one-to-one (1-1).

Proof: (Part 1: Nonsingular $\implies$ One-to-One) Assume T is nonsingular. Let $T(\alpha) = T(\beta)$. Subtract to one side:

$$T(\alpha) - T(\beta) = \bar{0}' \implies T(\alpha - \beta) = \bar{0}'$$

Because T is nonsingular, any vector mapping to zero must be the zero vector:

$$\alpha - \beta = \bar{0} \implies \alpha = \beta$$

Thus, T is one-to-one.

(Part 2: One-to-One $\implies$ Nonsingular) Assume T is one-to-one. We know from linear properties that $T(\bar{0}) = \bar{0}'$. Suppose there is an α such that $T(\alpha) = \bar{0}'$. Equating the two:

$$T(\alpha) = T(\bar{0})$$

Because T is one-to-one, equal images require equal inputs:

$$\alpha = \bar{0}$$

Thus, T is nonsingular. Both directions are proven.

22.4 3. Product (Composition) of Linear Transformations

22.4.1 Theorem 3: Composition is Linear

Statement: If $T : U \to V$ and $S : V \to W$ are two linear transformations, then their composition $ST : U \to W$ is also a linear transformation.

Proof: Let $a, b \in F$ and $\alpha, \beta \in U$. We evaluate the composition on a linear combination:

$$(ST)(a\alpha + b\beta) = S(T(a\alpha + b\beta))$$

Because T is linear:

$$= S(aT(\alpha) + bT(\beta))$$

Because S is linear:

$$= aS(T(\alpha)) + bS(T(\beta))$$

Rewriting using composition notation:

$$= a(ST)(\alpha) + b(ST)(\beta)$$

Thus, the composition ST is a linear transformation.

22.5 Key Takeaways

- **Invertible = Bijective**. A transformation can only be inverted if it is both one-to-one and onto.
- **Nonsingular = One-to-One**. A transformation is nonsingular if its null space only contains the zero vector. By Theorem 2, this guarantees the transformation is one-to-one.
- The inverse of a linear mapping is always linear, and the composition of two linear mappings is always linear.

22.6 What Comes Next

We will apply these theorems alongside the Rank-Nullity Theorem to solve numerical problems involving vector spaces.

23 Lecture 23: Vector Space - Numerical Questions on Rank-Nullity Theorem

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 23 of 27 **Video**: https://www.youtube.com/watch?v=3HoSPQmpzKc

23.1 Prerequisites

Before tackling numerical problems, we recall two quick but powerful theorems regarding linear transformations $T : U \to V$: 1. **Invertibility**: T is invertible if and only if T is non-singular. (A direct consequence of our previous theorems). 2. **Generators of Range**: If a set of vectors

$$S = \{u_1, u_2, \ldots, u_n\}$$

spans the domain U, then their images under T, which are

$$\{T(u_1), T(u_2), \ldots, T(u_n)\}$$

, will span the range space $R(T)$.

We will use the second theorem extensively: to find the range space of a transformation, we simply find the images of the standard basis vectors of the domain.

23.2 Numerical Example 1

Problem Statement: Show that the transformation

$$T : \mathbb{R}^2 \to \mathbb{R}^3$$

defined by:

$$T(a, b) = (a + b, a - b, b)$$

is a linear transformation. Then find its range space, rank, null space, and nullity, and verify the Rank-Nullity Theorem.

23.2.1 1. Prove Linearity

Let

$$\alpha = (a_1, b_1)$$

and

$$\beta = (a_2, b_2)$$

be vectors in $\mathbb{R}^2$, and let $x, y \in \mathbb{R}$ be scalars. Evaluate $T(x\alpha + y\beta)$:

$$T(x\alpha + y\beta) = T(xa_1 + ya_2, xb_1 + yb_2)$$

Apply the definition of T:

$$= ((xa_1 + ya_2) + (xb_1 + yb_2), (xa_1 + ya_2) - (xb_1 + yb_2), xb_1 + yb_2)$$

Group the x terms and y terms together:

$$= (x(a_1 + b_1) + y(a_2 + b_2), x(a_1 - b_1) + y(a_2 - b_2), xb_1 + yb_2)$$

Split the tuple:

$$\begin{aligned} &= x(a_1 + b_1, a_1 - b_1, b_1) + y(a_2 + b_2, a_2 - b_2, b_2) \\ &= xT(\alpha) + yT(\beta) \end{aligned}$$

Thus, T is a linear transformation.

23.2.2 2. Range Space and Rank

The domain is $\mathbb{R}^2$. Its standard basis is

$$S = \{e_1(1,0), e_2(0,1)\}$$

. Let's find their images: * $T(1,0) = (1+0, 1-0, 0) = (1,1,0)$ * $T(0,1) = (0+1, 0-1, 1) = (1,-1,1)$
These two vectors generate the range space $R(T)$:

$$R(T) = \{c_1(1,1,0) + c_2(1,-1,1) \mid c_1, c_2 \in \mathbb{R}\}$$

Since $(1,1,0)$ and $(1,-1,1)$ are not scalar multiples of each other, they are linearly independent and thus form a basis for $R(T)$. * **Rank** $\rho(T) =$ Dimension of $R(T) = \mathbf{2}$

23.2.3 3. Null Space and Nullity

The null space contains all vectors (a,b) that map to $(0,0,0)$:

$$T(a,b) = (0,0,0) \implies (a+b, a-b, b) = (0,0,0)$$

This gives the system: 1. $a+b=0$ 2. $a-b=0$ 3. $b=0$
From equation 3, $b=0$. Substituting into 1 gives $a=0$. The only solution is the zero vector.

$$N(T) = \{(0,0)\}$$

Because the null space contains only the zero vector, its basis is empty. * **Nullity** $\eta(T) =$ Dimension of $N(T)$ $= \mathbf{0}$

23.2.4 4. Verification

$$\text{Rank}(T) + \text{Nullity}(T) = 2 + 0 = 2$$

The dimension of the domain $\mathbb{R}^2$ is 2. The theorem holds perfectly.

23.3 Numerical Example 2

Problem Statement: Verify the Rank-Nullity Theorem for the linear transformation

$$T : \mathbb{R}^3 \to \mathbb{R}^3$$

defined by its action on the standard basis $\{e_1, e_2, e_3\}$:

$$T(e_1) = e_1 - e_2, \quad T(e_2) = 2e_2 + e_3, \quad T(e_3) = e_1 + e_2 + e_3$$

23.3.1 1. Range Space and Rank

Write the given images as coordinate vectors: *

$$T(e_1) = (1,-1,0)$$

- $$T(e_2) = (0,2,1)$$

- $$T(e_3) = (1,1,1)$$

These vectors span $R(T)$. However, observe that:

$$T(e_1) + T(e_2) = (1,-1,0) + (0,2,1) = (1,1,1) = T(e_3)$$

Because $T(e_3)$ is a linear combination of the other two, it is linearly dependent. We discard it from the basis. The remaining set $\{(1,-1,0),(0,2,1)\}$ is linearly independent and forms the basis. * **Rank** $\rho(T) = \mathbf{2}$

23.3.2 2. Null Space and Nullity

Let

$$\alpha = (x, y, z) = xe_1 + ye_2 + ze_3 \in N(T).$$

$$T(\alpha) = (0, 0, 0) \implies xT(e_1) + yT(e_2) + zT(e_3) = (0, 0, 0)$$

$$x(1, -1, 0) + y(0, 2, 1) + z(1, 1, 1) = (0, 0, 0)$$

$$(x + z, -x + 2y + z, y + z) = (0, 0, 0)$$

This gives the system: 1. $x + z = 0 \implies x = -z$ 2. $y + z = 0 \implies y = -z$ 3. $-x + 2y + z = 0 \implies -(-z) + 2(-z) + z = z - 2z + z = 0$ (Consistent)

Any vector in the null space has the form $(-z, -z, z) = z(-1, -1, 1)$. The null space is spanned by a single vector $\{(-1, -1, 1)\}$. * **Nullity $\eta(T) = 1$**

23.3.3 3. Verification

$$\text{Rank}(T) + \text{Nullity}(T) = 2 + 1 = 3$$

The dimension of the domain $\mathbb{R}^3$ is 3. The theorem holds.

23.4 Numerical Example 3

Problem Statement: Let T be a linear operator on $\mathbb{R}^3$ defined by:

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3)$$

Show that T is non-singular and invertible, and find a formula for T^{-1}.

23.4.1 1. Prove Non-Singularity

Set

$$T(x_1, x_2, x_3) = (0, 0, 0)$$

to find the null space. This creates a homogeneous system $AX = 0$:

$$A = \begin{pmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{pmatrix}$$

Find the determinant of the coefficient matrix A:

$$\det(A) = 3(4 - 0) - 0 + 1(-4 - (-1)) = 12 - 3 = 9$$

Since $\det(A) = 9 \neq 0$, the matrix is non-singular, meaning the homogeneous system has only the trivial solution

$$x_1 = 0, x_2 = 0, x_3 = 0.$$

Thus, $N(T) = \{(0, 0, 0)\}$, proving that T is non-singular. By our theorem, since it is an operator on a finite-dimensional space and is non-singular, it is invertible.

23.4.2 2. Find Inverse Formula

Set up the inverse relation:

$$T^{-1}(x_1, x_2, x_3) = (y_1, y_2, y_3) \iff T(y_1, y_2, y_3) = (x_1, x_2, x_3).$$

Substitute (y_1, y_2, y_3) into the definition of T: 1.

$$3y_1 + y_3 = x_1 \implies y_3 = x_1 - 3y_1$$

2.

$$-2y_1 + y_2 = x_2 \implies y_2 = x_2 + 2y_1$$

3.

$$-y_1 + 2y_2 + 4y_3 = x_3$$

Substitute (1) and (2) into (3):

$$-y_1 + 2(x_2 + 2y_1) + 4(x_1 - 3y_1) = x_3$$

$$-y_1 + 2x_2 + 4y_1 + 4x_1 - 12y_1 = x_3$$

$$-9y_1 + 4x_1 + 2x_2 = x_3 \implies 9y_1 = 4x_1 + 2x_2 - x_3 \implies \mathbf{y_1 = \frac{1}{9}(4x_1 + 2x_2 - x_3)}$$

Substitute y_1 back to find y_2:

$$y_2 = x_2 + 2y_1 = x_2 + \frac{2}{9}(4x_1 + 2x_2 - x_3) \implies \mathbf{y_2 = \frac{1}{9}(8x_1 + 13x_2 - 2x_3)}$$

Substitute y_1 back to find y_3:

$$y_3 = x_1 - 3y_1 = x_1 - \frac{3}{9}(4x_1 + 2x_2 - x_3) \implies \mathbf{y_3 = \frac{1}{9}(-3x_1 - 6x_2 + 3x_3)}$$

Thus, the final formula for the inverse mapping is:

$$T^{-1}(x_1, x_2, x_3) = \frac{1}{9}(4x_1 + 2x_2 - x_3, 8x_1 + 13x_2 - 2x_3, -3x_1 - 6x_2 + 3x_3)$$

24 Lecture 24: Vector Space - Eigenvalue & Eigenvector Theory

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 24 of 27 **Video**: https://www.youtube.com/watch?v=TH2nvOymftc

24.1 Prerequisites

We now turn our attention from transformations between different vector spaces to transformations from a vector space to itself ($T : V \to V$). These are called **linear operators**. When a linear operator acts on certain special vectors, it doesn't rotate or skew them; it simply scales them. These special vectors are called eigenvectors.

24.2 1. Fundamental Definitions

Let $T : V \to V$ be a linear transformation on an n-dimensional vector space V over a field F.

24.2.1 Eigenvalue and Eigenvector

A scalar $\lambda \in F$ is called an **eigenvalue** (or characteristic value) of T if there exists a **non-zero vector** $\alpha \in V$ ($\alpha \neq \bar{0}$) such that:

$$T(\alpha) = \lambda\alpha$$

The non-zero vector α is called the **eigenvector** (or characteristic vector) of T corresponding to the eigenvalue λ. *(Note: While the eigenvalue λ can be zero, the eigenvector α MUST NEVER be the zero vector.)*

24.2.2 Eigenspace

A single eigenvalue can have multiple eigenvectors associated with it. The set of all eigenvectors corresponding to an eigenvalue λ, along with the zero vector $\bar{0}$, is called the **Eigenspace** of λ.

$$W_\lambda = \{\alpha \in V \mid T(\alpha) = \lambda\alpha\}$$

24.2.3 Spectrum

The set of *all* distinct eigenvalues of the linear transformation T is called the **Spectrum** of T.

24.3 2. Fundamental Theorems

24.3.1 Theorem 1: Scalar Multiples of Eigenvectors

Statement: If α is an eigenvector of T corresponding to the eigenvalue λ, then any non-zero scalar multiple $k\alpha$ ($k \in F, k \neq 0$) is also an eigenvector of T corresponding to the same eigenvalue λ.
Proof: Since α is an eigenvector for λ:

$$T(\alpha) = \lambda\alpha \quad (\alpha \neq \bar{0})$$

Let $k \in F$ with $k \neq 0$. Because $\alpha \neq \bar{0}$ and $k \neq 0$, the scalar multiple $k\alpha \neq \bar{0}$. Let's find the image of $k\alpha$:

$$T(k\alpha) = kT(\alpha) \quad \text{(since } T \text{ is linear)}$$

Substitute $T(\alpha) = \lambda\alpha$:

$$= k(\lambda\alpha)$$

By associativity and commutativity of scalars in the field F:

$$= (k\lambda)\alpha = (\lambda k)\alpha = \lambda(k\alpha)$$

Thus, $T(k\alpha) = \lambda(k\alpha)$. Since $k\alpha$ is non-zero, it is an eigenvector corresponding to λ.

24.3.2 Theorem 2: Uniqueness of Eigenvalues

Statement: A single eigenvector α cannot correspond to more than one distinct eigenvalue.

Proof: Let us assume the contrary. Suppose the eigenvector α corresponds to two distinct eigenvalues, λ_1 and λ_2 ($\lambda_1 \neq \lambda_2$). By definition, we have: 1.

$$T(\alpha) = \lambda_1 \alpha$$

2.

$$T(\alpha) = \lambda_2 \alpha$$

Because the mapping T is a function, a single input α must have a unique output. Therefore, we can equate the right-hand sides:

$$\lambda_1 \alpha = \lambda_2 \alpha$$

Bring all terms to one side:

$$\lambda_1 \alpha - \lambda_2 \alpha = \bar{0} \implies (\lambda_1 - \lambda_2)\alpha = \bar{0}$$

By definition, an eigenvector α cannot be the zero vector ($\alpha \neq \bar{0}$). In a vector space, if a scalar times a non-zero vector equals zero, the scalar itself must be zero.

$$\lambda_1 - \lambda_2 = 0 \implies \lambda_1 = \lambda_2$$

This contradicts our assumption that λ_1 and λ_2 are distinct. Therefore, an eigenvector corresponds to exactly one eigenvalue.

24.3.3 Theorem 3: The Eigenspace is a Subspace

Statement: The eigenspace W_λ corresponding to an eigenvalue λ of a linear transformation $T : V \to V$ is a subspace of the vector space V.

Proof: By definition,

$$W_\lambda = \{\alpha \in V \mid T(\alpha) = \lambda\alpha\}.$$

First, note that the zero vector $\bar{0} \in V$ satisfies $T(\bar{0}) = \bar{0} = \lambda\bar{0}$. Thus, $\bar{0} \in W_\lambda$, meaning W_λ is non-empty.

To prove it is a subspace, we must show it is closed under linear combinations. Let $\alpha, \beta \in W_\lambda$ and let $a, b \in F$ be any scalars. Since $\alpha, \beta \in W_\lambda$: 1. $T(\alpha) = \lambda\alpha$ 2. $T(\beta) = \lambda\beta$

Let's evaluate the transformation on the linear combination $a\alpha + b\beta$:

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) \quad \text{(since } T \text{ is linear)}$$

Substitute the equations from above:

$$= a(\lambda\alpha) + b(\lambda\beta)$$

Reorder the scalars (since F is a field):

$$= \lambda(a\alpha) + \lambda(b\beta)$$

Factor out λ:

$$= \lambda(a\alpha + b\beta)$$

Since $T(a\alpha + b\beta) = \lambda(a\alpha + b\beta)$, the vector $(a\alpha + b\beta)$ satisfies the condition to be in W_λ. Therefore, W_λ is closed under linear combinations and is a valid subspace of $V(F)$.

24.4 Key Takeaways

- **Eigenvalues** scale **Eigenvectors** without changing their direction.
- The set of all eigenvectors for a given eigenvalue forms a complete, well-behaved vector subspace called the **Eigenspace**.
- A single eigenvalue can have infinitely many eigenvectors (Theorem 1 and 3), but a single eigenvector is tied exclusively to one eigenvalue (Theorem 2).

25 Lecture 25: Vector Space - Numerical Problem on Eigen Value & Vectors

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 25 of 27 **Video**: https://www.youtube.com/watch?v=CDKS5Se6r3Q

25.1 Prerequisites

Before moving to numerical problems and diagonalization, we must finish the theoretical foundation of eigenvalues. Today we will prove that distinct eigenvalues produce fundamentally unique (linearly independent) eigenvectors, and we will establish the characteristic equation used to calculate eigenvalues.

25.2 Theorem 4: Independence of Distinct Eigenvectors

Statement: Distinct non-zero eigenvectors of a linear operator T corresponding to distinct eigenvalues of T are linearly independent.

Proof (by Mathematical Induction): Let $T : V \to V$ be a linear operator. Let

$$\alpha_1, \alpha_2, \ldots, \alpha_n$$

be eigenvectors corresponding to distinct eigenvalues

$$\lambda_1, \lambda_2, \ldots, \lambda_n.$$

This means:

$$T(\alpha_i) = \lambda_i \alpha_i$$

for all $i = 1, \ldots, n$. We want to prove that the set

$$S_n = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

is linearly independent. We proceed by induction on n.

Base Case $(n = 1)$: For $n = 1$, the set is

$$S_1 = \{\alpha_1\}.$$

Because α_1 is an eigenvector, it is a non-zero vector $(\alpha_1 \neq \bar{0})$. Any set containing a single non-zero vector is linearly independent. The base case holds.

Induction Hypothesis $(n = k)$: Assume the theorem holds for k eigenvectors. Thus, the set

$$S_k = \{\alpha_1, \alpha_2, \ldots, \alpha_k\}$$

is linearly independent.

Inductive Step $(n = k + 1)$: We must prove that

$$S_{k+1} = \{\alpha_1, \ldots, \alpha_k, \alpha_{k+1}\}$$

is linearly independent. Let's set up a linear combination equal to the zero vector:

$$a_1\alpha_1 + a_2\alpha_2 + \cdots + a_k\alpha_k + a_{k+1}\alpha_{k+1} = \bar{0} \quad \text{— (1)}$$

Apply the linear operator T to both sides. Since $T(\bar{0}) = \bar{0}$:

$$a_1T(\alpha_1) + a_2T(\alpha_2) + \cdots + a_kT(\alpha_k) + a_{k+1}T(\alpha_{k+1}) = \bar{0}$$

Substitute

$$T(\alpha_i) = \lambda_i\alpha_i :$$

$$a_1\lambda_1\alpha_1 + a_2\lambda_2\alpha_2 + \cdots + a_k\lambda_k\alpha_k + a_{k+1}\lambda_{k+1}\alpha_{k+1} = \bar{0} \quad \text{— (2)}$$

Now, multiply the original equation (1) by the scalar λ_{k+1}:

$$a_1\lambda_{k+1}\alpha_1 + a_2\lambda_{k+1}\alpha_2 + \cdots + a_k\lambda_{k+1}\alpha_k + a_{k+1}\lambda_{k+1}\alpha_{k+1} = \bar{0} \quad \text{— (3)}$$

Subtract equation (3) from equation (2). The last term (a_{k+1}) perfectly cancels out:

$$a_1(\lambda_1 - \lambda_{k+1})\alpha_1 + a_2(\lambda_2 - \lambda_{k+1})\alpha_2 + \cdots + a_k(\lambda_k - \lambda_{k+1})\alpha_k = \bar{0} \quad \text{— (4)}$$

Look at equation (4). It is a linear combination of the first k eigenvectors equal to zero. By our induction hypothesis, the first k eigenvectors are linearly independent. Therefore, every single scalar coefficient in this equation must be exactly zero:

$$a_i(\lambda_i - \lambda_{k+1}) = 0 \quad \text{for all } i = 1, \ldots, k$$

Because the eigenvalues are distinct, $\lambda_i \neq \lambda_{k+1}$, meaning

$$(\lambda_i - \lambda_{k+1}) \neq 0$$

. The only way the product can be zero is if:

$$a_1 = 0, \quad a_2 = 0, \quad \ldots, \quad a_k = 0$$

Substitute these zero coefficients back into equation (1):

$$0 \cdot \alpha_1 + \cdots + 0 \cdot \alpha_k + a_{k+1}\alpha_{k+1} = \bar{0} \implies a_{k+1}\alpha_{k+1} = \bar{0}$$

Because α_{k+1} is an eigenvector, it is not zero ($\alpha_{k+1} \neq \bar{0}$). Thus,

$$a_{k+1} = 0.$$

Since

$$a_1 = a_2 = \cdots = a_{k+1} = 0$$

, the set S_{k+1} is linearly independent. By mathematical induction, the theorem holds for any number n.

25.2.1 Corollary: Maximum Number of Distinct Eigenvalues

Statement: A linear operator T on an n-dimensional vector space V can have at most n distinct eigenvalues.

Proof: Assume, for contradiction, that T has $n+1$ distinct eigenvalues. By Theorem 4, their corresponding $n+1$ eigenvectors must form a linearly independent set in V. However, it is a fundamental property of vector spaces that an n-dimensional space can contain a maximum of n linearly independent vectors. The existence of $n+1$ linearly independent vectors is a contradiction. Thus, T cannot have more than n distinct eigenvalues.

25.3 Theorem 5: The Characteristic Equation

Statement: Let T be a linear operator on a finite-dimensional vector space V. The following three statements are mathematically equivalent: 1. λ is an eigenvalue of T. 2. The operator $(T - \lambda I)$ is singular. 3. $\det(T - \lambda I) = 0$.

Proof (Circular Implication):

Part 1: (1) $\implies$ (2) Assume λ is an eigenvalue of T. By definition, there is a non-zero vector $\alpha \in V$ such that $T(\alpha) = \lambda\alpha$. Rewrite this:

$$T(\alpha) - \lambda\alpha = \bar{0}$$

$$T(\alpha) - \lambda I(\alpha) = \bar{0} \implies (T - \lambda I)(\alpha) = \bar{0}$$

Since the operator $(T - \lambda I)$ maps a non-zero vector α to the zero vector, its null space is not trivial. Therefore, $(T - \lambda I)$ is a singular operator.

Part 2: (2) $\implies$ (3) Assume $(T - \lambda I)$ is singular. In a finite-dimensional space, a singular operator maps multiple vectors to zero, meaning its rank is less than the dimension of the space. Because it does not have full rank, the determinant of its matrix representation must be zero:

$$\det(T - \lambda I) = 0$$

Part 3: (3) $\implies$ (1) Assume $\det(T - \lambda I) = 0$. Because the determinant is zero, the matrix is not invertible, which means the operator $(T - \lambda I)$ is singular. Because it is singular, its null space contains at least one non-zero vector. Let this non-zero vector be α.

$$(T - \lambda I)(\alpha) = \bar{0}$$

Expand this:

$$T(\alpha) - \lambda I(\alpha) = \bar{0} \implies T(\alpha) = \lambda\alpha$$

Because α is a non-zero vector that satisfies this relation, λ is, by definition, an eigenvalue of T. The equivalence of all three statements is proven.

25.4 Key Takeaways

- **Distinct Eigenvalues $\implies$ Independence**: Eigenvectors from different eigenvalues point in fundamentally different, non-overlapping directions in the vector space.
- **The Characteristic Equation**: Theorem 5 provides the computational method to find eigenvalues. To find the eigenvalues of a transformation (or matrix), you set $\det(T - \lambda I) = 0$ and solve for λ. This is called the characteristic equation.

25.5 What Comes Next

We will use the characteristic equation $\det(A - \lambda I) = 0$ to calculate the eigenvalues of matrices and determine if those matrices can be diagonalized.

26 Lecture 26: Vector Space - Numerical Problems on Eigenvalues & Eigenvectors of Matrices

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 26 of 27 **Video**: https://www.youtube.com/watch?v=X7tLRj8Le-8

26.1 Prerequisites

To find the eigenvalues and eigenvectors of a linear transformation, we can find the eigenvalues and eigenvectors of its matrix representation. Today, we focus purely on the computational mechanics of extracting these values from square matrices. This process is the foundational step for matrix diagonalization.

26.2 The General Process

Let A be a square matrix. 1. **Find Eigenvalues**: Solve the characteristic equation for λ:

$$\det(A - \lambda I) = 0$$

`This yields a polynomial (quadratic for $2 \times 2$, cubic for $3 \times 3$). The roots of this polynom`

2. **Find Eigenvectors**: For each eigenvalue λ, solve the homogeneous system of linear equations:

$$(A - \lambda I)X = \bar{0}$$

`Reduce the matrix $(A - \lambda I)$ to row echelon form to find the free variables. Assign values to th`

26.3 Problem 1: Matrix (Distinct Eigenvalues)

Given:

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$$

Step 1: Find Eigenvalues Set up the characteristic equation:

$$\det \begin{bmatrix} 4-\lambda & 2 \\ 3 & 3-\lambda \end{bmatrix} = 0$$

Expand the determinant:

$$(4-\lambda)(3-\lambda) - (3)(2) = 0$$

$$\lambda^2 - 7\lambda + 12 - 6 = 0 \implies \lambda^2 - 7\lambda + 6 = 0$$

Factor the quadratic equation:

$$(\lambda - 6)(\lambda - 1) = 0$$

The eigenvalues are **

$$\lambda_1 = 1$$

** and **

$$\lambda_2 = 6$$

**.

Step 2: Find Eigenvector for $\lambda = 1$

$$(A - 1I)X = \bar{0} \implies \begin{bmatrix} 3 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Apply row operation $R_2 \to R_2 - R_1$:

$$\begin{bmatrix} 3 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 3x_1 + 2x_2 = 0$$

We have 1 free variable. Choose

$$x_1 = 2 \implies x_2 = -3.$$

The eigenvector and its eigenspace:

$$X^{(1)} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad W_1 = \text{span}\left\{\begin{bmatrix} 2 \\ -3 \end{bmatrix}\right\}$$

Step 3: Find Eigenvector for $\lambda = 6$

$$(A - 6I)X = \bar{0} \implies \begin{bmatrix} -2 & 2 \\ 3 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Apply row operation $R_2 \to 2R_2 + 3R_1$:

$$\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -2x_1 + 2x_2 = 0 \implies x_1 = x_2$$

Choose

$$x_2 = 1 \implies x_1 = 1.$$

The eigenvector and its eigenspace:

$$X^{(2)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad W_6 = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$$

26.4 Problem 2: Matrix (Repeated Eigenvalues)

Given:

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Step 1: Find Eigenvalues

$$\det \begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} = 0$$

Expanding along the first row and simplifying yields the characteristic polynomial:

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

Using trial-and-error, we test $\lambda = 2$:

$$2^3 - 12(2)^2 + 36(2) - 32 = 8 - 48 + 72 - 32 = 0$$

Since $\lambda = 2$ is a root, we factor out $(\lambda - 2)$:

$$(\lambda - 2)(\lambda^2 - 10\lambda + 16) = 0 \implies (\lambda - 2)(\lambda - 2)(\lambda - 8) = 0$$

The eigenvalues are $\lambda = 2, 2, 8$. Notice that 2 is a repeated eigenvalue.

Step 2: Find Eigenvector for $\lambda = 8$

$$(A - 8I)X = \bar{0} \implies \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Reduce to row echelon form:

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives equations: 1.

$$-x_1 - x_2 + x_3 = 0$$

2.

$$-3x_2 - 3x_3 = 0 \implies x_2 = -x_3$$

Choose

$$x_3 = 1 \implies x_2 = -1 \implies x_1 = 2.$$

$$X^{(1)} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Step 3: Find Eigenvectors for the Repeated Eigenvalue $\lambda = 2$ Because $\lambda = 2$ appears twice (algebraic multiplicity of 2), we anticipate finding 2 free variables (geometric multiplicity of 2).

$$(A - 2I)X = \bar{0} \implies \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Notice that row 2 and row 3 are scalar multiples of row 1. Reducing the matrix zeros out the bottom two rows entirely:

$$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies 2x_1 - x_2 + x_3 = 0$$

We have 3 variables and only 1 equation, meaning we have $3 - 1 = 2$ free parameters. We make two separate sets of choices to find two linearly independent eigenvectors:

- **Choice A:** Let

$$x_3 = 1$$

and

$$x_2 = -1.$$

$$2x_1 - (-1) + 1 = 0 \implies 2x_1 = -2 \implies x_1 = -1$$

$$X^{(2)} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

- **Choice B:** Let

$$x_3 = 0$$

and

$$x_2 = 2.$$

$$2x_1 - (2) + 0 = 0 \implies 2x_1 = 2 \implies x_1 = 1$$

$$X^{(3)} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

The eigenspace for $\lambda = 2$ is the span of these two vectors:

$$W_2 = \text{span} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \right\}$$

26.5 Key Takeaways

- The characteristic polynomial for an $n \times n$ matrix is always of degree n, yielding exactly n eigenvalues (some may be repeated or complex).
- For repeated eigenvalues, if the algebraic multiplicity (how many times the root appears) equals the geometric multiplicity (the number of free variables / linearly independent eigenvectors found), the matrix is well-behaved and diagonalizable.
- Row reduction is strictly required to find eigenvectors correctly.

26.6 What Comes Next

We will use these matrices of eigenvectors to perform **Diagonalization**, simplifying complex transformations into pure scaling operations.

27 Lecture 27: Vector Space - Diagonalization of Matrices

Series: Vector Space by Bhagwan Singh Vishwakarma **Lecture**: 27 of 27 **Video**: https://www.youtube.com/watch?v=qMwnGmEXBpY

27.1 Prerequisites

Diagonalization is the final culmination of our study of eigenvalues and eigenvectors. To prove that a matrix is diagonalizable means to prove that there exists a change of basis (a transition matrix) that transforms the given matrix into a purely diagonal matrix. This greatly simplifies matrix computations, especially matrix exponentiation.

27.2 The Working Procedure for Diagonalization

Suppose we want to determine if an $n \times n$ square matrix A is diagonalizable, and if so, perform the diagonalization.

1. **Characteristic Equation**: Solve $\det(A - \lambda I) = 0$ to find the n eigenvalues
$$\lambda_1, \lambda_2, \ldots, \lambda_n.$$
2. **Eigenvectors**: For each eigenvalue, solve $(A - \lambda I)X = \bar{0}$ to find the corresponding eigenvectors. We must be able to generate a complete set of n **linearly independent** eigenvectors. If we cannot find n linearly independent eigenvectors, the matrix is *not* diagonalizable.
3. **Transition (Modal) Matrix (P)**: Construct the transition matrix P by placing the n linearly independent eigenvectors as its columns:
$$P = \begin{bmatrix} X_1 & X_2 & \ldots & X_n \end{bmatrix}$$
4. **Inverse Modal Matrix (P^{-1})**: Compute the inverse of the transition matrix. For smaller matrices, use the adjoint method:
$$P^{-1} = \frac{1}{\det(P)}\text{adj}(P)$$
5. **Compute $P^{-1}AP$**: Multiply the matrices. If the matrix is diagonalizable, the result will exactly equal a diagonal matrix D, whose diagonal entries are the eigenvalues of A in the exact same order as their corresponding eigenvectors appear in P.
$$P^{-1}AP = D = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix}$$

27.3 Example 1: Diagonalizing a Matrix

Given:
$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$$

- Eigenvalues:
$$\lambda_1 = 1,$$
$$\lambda_2 = 6.$$
- Eigenvectors:

$$X_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 1: Construct the Modal Matrix P Place X_1 and X_2 as columns:

$$P = \begin{bmatrix} 2 & 1 \\ -3 & 1 \end{bmatrix}$$

Check linear independence by finding the determinant:

$$\det(P) = (2)(1) - (1)(-3) = 2 + 3 = 5 \neq 0$$

Since $\det(P) \neq 0$, the columns are linearly independent.

Step 2: Find P^{-1} Find the cofactors to build the adjoint matrix: *

$$P_{11} = 1, \quad P_{12} = 3$$

-

$$P_{21} = -1, \quad P_{22} = 2$$

$$\operatorname{adj}(P) = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix} \implies P^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

Step 3: Evaluate $P^{-1}AP$ First, multiply A and P:

$$AP = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} (8-6) & (4+2) \\ (6-9) & (3+3) \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ -3 & 6 \end{bmatrix}$$

Next, multiply P^{-1} by the result:

$$P^{-1}(AP) = \frac{1}{5}\begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 2 & 6 \\ -3 & 6 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} (2+3) & (6-6) \\ (6-6) & (18+12) \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 5 & 0 \\ 0 & 30 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix} = D$$

Because $P^{-1}AP$ yields a diagonal matrix containing the eigenvalues 1 and 6 on the diagonal, we have successfully proven that matrix A is diagonalizable.

27.4 Example 2: Matrix with Repeated Eigenvalues

Given:

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

Step 1: Eigenvalues and Eigenvectors * **Eigenvalues**: Solving $\det(A - \lambda I) = 0$ gives $\lambda = 8, -1, -1$. (Note that -1 is a repeated eigenvalue with algebraic multiplicity 2). * **Eigenvector for** $\lambda = 8$: Solving $(A - 8I)X = \bar{0}$ yields:

$$X_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

- **Eigenvectors for** $\lambda = -1$: Solving $(A + 1I)X = \bar{0}$ yields the single equation

$$2x_1 + x_2 + 2x_3 = 0$$

. By setting

$$(x_1 = 1, x_3 = 0)$$

and then

$$(x_1 = 0, x_3 = 1)$$

, we extract two linearly independent eigenvectors:

$$X_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Step 2: Modal Matrix P Place X_1, X_2, and X_3 into the modal matrix:

$$P = \begin{bmatrix} 2 & 1 & 0 \\ 1 & -2 & -2 \\ 2 & 0 & 1 \end{bmatrix}$$

Step 3: Diagonalization By computing P^{-1} using the adjoint method and evaluating the product $P^{-1}AP$, we get:

$$P^{-1}AP = \begin{bmatrix} 8 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = D$$

The eigenvalues $8, -1, -1$ perfectly match the order of the eigenvectors in P. The matrix A is diagonalizable.

27.5 Key Takeaways

- **Diagonalization represents a change of basis**: By shifting our perspective to the basis formed by the eigenvectors, the complex transformation A simplifies into a diagonal matrix D, which only scales vectors along these specific axes.
- **A requirement for diagonalization**: You *must* have a full set of n linearly independent eigenvectors for an $n \times n$ matrix to be diagonalizable.

(This concludes the 27-lecture Vector Space series!)